

Bootstrapping-Free Blind PCS: Achieving Constant Depth and Linear Prover

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Abstract. In this work, we construct a new and highly efficient blind polynomial commitment scheme (PCS) over non-binary fields. Our scheme is specifically designed to handle encrypted coefficients without requiring expensive bootstrapping operations, achieving a breakthrough in the "complexity-depth" trade-off.

The proposed scheme features an extremely efficient prover both asymptotically and concretely. The commitment and evaluation phases are dominated by a strictly linear $O(n)$ number of field operations. Furthermore, the construction maintains a constant multiplicative depth, which is a critical requirement for efficiency in homomorphic encryption settings. Concretely, for large-scale circuit sizes, our prover is significantly faster than prior state-of-the-art schemes such as *Phalanx* and *LAMINATE*.

Our underlying technique is the Generalized RAA code, an extremely efficient error-correcting code that extends the binary RAA code structure to arbitrary non-binary prime fields $\mathbb{F}_p$. We analyze the bounds over non-binary fields, which demonstrate that this code maintains a linear minimum distance property with high probability. By combining Ligeró's IOPP framework, we obtain the first asymptotically and concretely good blind PCS that achieves strictly linear $O(n)$ encoding complexity for the prover while avoiding the expensive bootstrapping operations.

Keywords: VCoED · Blind SNARKs · RAA Code · PCS

1 Introduction

Verifiable Computation over Encrypted Data (VCoED)[3,15,7,18] enables users to outsource complex computations to untrusted servers while ensuring both strict data privacy and the integrity of the results. The primary vehicle for realizing VCoED is the Blind SNARK[17,26,32], which essentially adapts traditional SNARKs to FHE ciphertexts. This allows a prover to generate succinct proofs for statements regarding the underlying plaintexts without requiring decryption. Following the standard SNARK paradigm[16,11,28,13,25], a blind SNARK is typically instantiated by combining a blind Polynomial Interactive Oracle Proof (PIOP) with a blind Polynomial Commitment Scheme (PCS), variants of

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the traditional PIOP and PCS that operates directly on encrypted coefficients. Targeting the application of blind SNARKs to VCoED, recent state-of-the-art works, such as *Phalanx*[32] and *LAMINATE*[26], have achieved remarkable efficiency gains by designing FHE-efficient PIOP. These systems optimize prover performance by specializing the PIOP layer to exploit the SIMD-parallelism of underlying FHE schemes like BGV[8] or BFV[9].

A Blind PCS is a cryptographic primitive that enables a prover to commit to an encrypted polynomial and provide succinct proofs of its encrypted evaluations without accessing the plaintexts. Unlike traditional PCS, a Blind PCS must operate within the strict noise budget of the FHE scheme[9,8,14,12]. Excessive multiplicative depth during the commitment or evaluation phases can lead to decryption failure, necessitating prohibitively expensive **bootstrapping** operations[19,9,8,12]. As explicitly identified in *LAMINATE*, there is currently no existing Blind PCS construction that can simultaneously achieve constant multiplicative depth and strictly linear (or even quasilinear) prover complexity.

Every existing Blind PCS[3,18,32,26] construction fundamentally relies on an underlying Error-Correcting Code (ECC) as the primary mechanism for proximity testing, which ensures that the committed vector is indeed close to the space of codewords. However, current ECC adopted in blind PCS cannot simultaneously achieve strictly linear-time encoding and constant multiplicative depth. Take Reed-Solomon (RS) code[29], the most widely adopted ECC in SNARKs, as a primary example. The standard implementation of RS code via the Number Theoretic Transform (NTT) introduces a severe "complexity-depth" dilemma in the VCoED setting. An alternative k -depth 2D-NTT approaches force a prohibitive escalation in prover complexity to $O(kn^{1+1/k})$. This unresolved "complexity-depth" dilemma motivates our central research question:

What is the optimal asymptotic complexity achievable for a blind PCS with a constant multiplicative depth?

1.1 Our Contributions

In this paper, we demonstrate that it is possible to achieve strictly linear $O(N)$ prover complexity while maintaining a constant multiplicative depth. We provide a solution through the construction of the Generalized Repeat-Accumulate-Accumulate (RAA) code over large fields $\mathbb{F}_t$. The code can achieve the linear minimum distance properties of the RAA structure for arbitrary q , while maintaining the linear time encoding property. Based on the code, we construct the first FHE-friendly PCS that satisfies all three desiderata: strictly linear prover time, constant multiplicative depth, and native field alignment with BGV/BFV plaintext spaces.

To this end, our work delivers the following advancements:

Bounds for Generalized RAA Codes. We provide a rigorous analysis of the minimum distance properties of RAA codes across non-binary fields. While

standard RAA codes are traditionally defined over $\mathbb{F}_2$, we extend their combinatorial structure to large prime fields $\mathbb{F}_t$, which are natively used as FHE plaintext spaces. We derive new theoretical bounds for the minimum distance of these non-binary variants. This analysis ensures that our code maintains the high soundness error required for secure commitment schemes while operating in an FHE-friendly domain.

Theorem 1. *Denote S_N as the symmetric group of degree N i.e. the collection of all permutations on $\{1, 2, \dots, N\}$. Denote k as the repeat times of the repeat code. There exists a constant $K \in \mathbb{N}^+$ such that for any $k > K$, the following holds:*

$$\Pr_{\substack{\pi_1, \pi_2 \xleftarrow{\$} S_N \\ \mathbf{r}_1, \mathbf{r}_2 \xleftarrow{\$} \mathbb{F}_p^N}} [\exists x \text{ s.t. } \text{wt}(\text{RAA}(x; \pi_1, \pi_2, \mathbf{r}_1, \mathbf{r}_2)) \leq \delta N] = o\left(\frac{1}{N}\right)$$

Construction of the first asymptotically and concretely good blind PCS

Leveraging the generalized RAA code, we construct the first Blind PCS that achieves strictly linear encoding complexity for the Prover. Unlike existing state-of-the-art schemes such as *Phalanx* and *LAMINATE*, our scheme utilizes a recursive accumulation process. This allows for constant multiplicative depth and eliminates the "complexity-depth" dilemma. In addition, we implement the engineering optimization that exploits the SIMD capabilities of BGV/BFV FHE schemes.

Theorem 2. *There exists a blind Polynomial Commitment Scheme with:*

- **Prover Complexity:** $O(N)$ plaintext-ciphertext multiplications.
- **Verifier Complexity:** $O(\lambda\sqrt{N})$ plaintext-plaintext multiplications and $O(\lambda\sqrt{N})$ decryptions.
- **Proof Size:** $O(\lambda\sqrt{N})$ ciphertexts and $O(\lambda \log N)$ hashes.
- **Multiplication depth on ciphertexts:** $O(1)$ multiplication depth.

In addition, the PCS has the following bound for the soundness error:

$$\epsilon_{total} \leq \left(1 - \frac{\delta}{3}\right)^\ell + \frac{N}{p^k}$$

where ℓ denotes the query times, δ denotes the relative minimum distance of Generalized RAA code and $\mathbb{F}_{p^k}$ is the extension field.

1.2 Technical Overview

Key Observation We observe that accumulator-based codes[23] have emerged as a prominent choice for the underlying ECC in several state-of-the-art PCS implementations, particularly those designed for strictly linear prover complexity and concrete efficiency in prover time. In the accumulator-based structures, the

input message c_0 is first reordered through a randomized permutation $c_1 = \pi(c_0)$, and subsequently processed by an accumulator A that computes the prefix sum of the permuted elements $c_2 = A(\pi(c_0))$. This specific design is important to our research objectives as it imposes minimal computational overhead and ensures a controllable multiplicative depth. Specifically, the accumulation process contributes only a single level of multiplicative depth, while the overall structure enables strictly linear-time encoding, facilitating a highly efficient prover.

However, the practical utility of these codes in the FHE setting is currently limited by several constraints. Most of the accumulator-based codes are designed for binary scenarios. Yet efficient SIMD operations in FHE typically require the plaintext domain to be an odd prime field $\mathbb{F}_q$, a prerequisite for batching multiple plaintexts into a single ciphertext. While Expander-Accumulate (EA) codes [6, 27] attempted to generalize this paradigm to larger fields, they introduce an $O(\log n)$ sparsity requirement for the expander graphs, resulting in a suboptimal $O(n \log n)$ complexity that fails the requirement for strict linearity.

Generalized RAA Code To overcome these problems, we extend the binary RAA structure to arbitrary non-binary fields by introducing randomized weighting vectors alongside permutations. The generalized RAA code decouples the weight distribution from the input message values, allowing us to rigorously prove that the code maintains a linear minimum distance property with high probability in large prime fields.

Blind PCS Construction via Ligerio IOPP Leveraging this code, we instantiate the first Bootstrapping-Free linear time blind PCS. Our construction integrates the Generalized RAA code into the Ligerio IOPP framework by replacing the traditional RS codes. By operating directly on encrypted coefficients with $O(1)$ multiplicative depth, our prover avoids the prohibitive noise growth of NTT-based commitments. Furthermore, we optimize the protocol using SIMD technique, grouping matrix rows within ciphertext slots to perform parallel homomorphic encoding and aggregation.

1.3 Organization

In section 2, we review the related works of coding theory, SNARKs and Blind SNARKs. In section 3, we introduce the preliminaries that are used in the following sections. In section 4, we introduce the construction of the generalized RAA Code and prove the asymptotically good distance property of it. In section 5, we provide the numerical analysis of some core functions mentioned in Section 4, aiming to offer intuitive insights to the readers. In addition, we provide concrete failure probabilities under different parameters. In Section 6, we develop and analyze a Blind PCS based on our Generalized RAA Code. In Section 7, we provide an empirical evaluation of the scheme, including a benchmark comparison with competitive baselines.

2 Related Works

While the development of Blind zkSNARKs has significantly advanced the field of VCoED [3,15,7], the majority of recent advancements have focused predominantly on optimizing the PIOP layer [32,26] rather than addressing the inherent inefficiencies of the commitment phase. Consequently, there remains a notable lack of innovation directed at the PCS specifically tailored for the blind setting. In this section, we survey the research landscape of blind PCS and coding based PCS.

2.1 Blind PCS

The development of Blind PCS is closely intertwined with the adaptation of interactive proofs to the FHE domain.

The concept of performing proximity testing over encrypted values can be traced back to the work [18]. They provided the first foundational study on applying the FRI (Fast Reed-Solomon IOPP) protocol to FHE ciphertexts. However, their security model remains relatively constrained, as their notion only considers the decryption of freshly encrypted ciphertexts that have not yet undergone homomorphic operations.

Concurrently, Aranha et al.[3] investigated the direct application of FRI to ciphertexts by compiling δ -correlated IOPs into SNARKs. Notably, their approach bypasses the explicit polynomial commitment abstraction. To address the noise constraints of FHE, they proposed the use of 2D-NTT techniques to achieve a constant-depth NTT, albeit at the cost of increased computational complexity. While Aranha et al.[3] successfully demonstrated a practical implementation of FRI over encrypted values, the FRI protocol within their framework still suffers from significant multiplicative depth.

To address this fundamental noise constraint, *Phalanx*[32] advanced the field by proposing the first constant-depth full SNARK for the VCoED setting. By shifting the focus toward the efficient execution of the sumcheck protocol over ciphertexts within the Spartan framework, *Phalanx* effectively decoupled the circuit multiplicative depth from the FHE noise growth. For its underlying PCS, *Phalanx* utilizes a combination of Ligerio and Reed-Solomon codes to handle encrypted polynomial commitments, optimizing for SIMD-parallelism in schemes like BGV or BFV. However, both its prover time and proof size escalate significantly as the circuit scale increases.

This efficiency gap motivated the development of *LAMINATE*[26], which addresses the proof size explosion by investigating novel methods to pack ciphertexts more effectively. By adapting the GKR protocol and incorporating folding-based techniques like blind Basefold, *LAMINATE* aims to optimize the communication complexity of VCoED systems without sacrificing the noise-efficiency of constant-depth provers.

However, despite these advancements in optimizing communication complexity and proof size, none of these works have bypassed the "complexity-depth" dilemma.

2.2 Coding-based PCS

Beyond the specialized domain of Blind PCS, we also examine the development of traditional PCS. We categorize the evolution of coding-based PCS into three primary categories:

RS Code based PCS [5,31,2,21,33,4] This family of constructions relies on the optimal distance properties of Reed-Solomon (RS) codes. The **FRI** protocol [5] introduced a recursive folding technique to achieve logarithmic proof size, which has been the standard for transparent SNARKs. Recent advancements like **Basefold** [31] have generalized this approach to be field-agnostic by utilizing a broader class of codes. However, as analyzed in our problem statement, any RS code-based construction inherently requires $O(n \log n)$ encoding complexity due to the necessity of NTTs. In the blind setting, the logarithmic multiplicative depth of NTTs (or the complexity explosion of constant-depth grouped NTTs) poses a significant bottleneck for FHE-based provers.

Linear-time Encodable Code based PCS [20,30] To achieve a strictly linear-time prover, **Brakedown** [20] replaced RS codes with linear-time encodable codes within a matrix-based commitment framework. This line of work culminated in **Orion** [30], which achieved $O(n)$ prover time and $O(\log^2 n)$ proof size. While these schemes are asymptotically optimal in the plaintext domain, they are not natively FHE-friendly. Their internal algebraic structures often involve complex bit-level operations or high-depth circuits that lead to unsustainable noise growth in BGV/BFV schemes, rendering them impractical for Blind SNARKs.

The Accumulator-based PCS [10,6] Another promising direction involves codes with a recursive accumulation structure. **Blaze** [10] and Alexander et al.[6] leverage the fact that accumulation (or prefix sums) can be computed in linear time. These constructions offer a balance between prover efficiency and structural simplicity. However, existing work in this category is primarily optimized for binary fields or specific cryptographic groups. There is a lack of research into non-binary variants that can natively align with the large prime fields $\mathbb{F}_t$ used in FHE, which is the gap our Generalized RAA-based PCS aims to fill.

In summary, while traditional PCS research has successfully yielded strictly linear-time encodable constructions, these schemes almost universally exhibit characteristics that are fundamentally incompatible with the constraints of Fully Homomorphic Encryption. Specifically, many state-of-the-art linear-time codes involve high-depth algebraic structures that lead to unsustainable noise growth, while others suffer from a "field conflict" where their binary-oriented logic cannot be efficiently mapped to the large prime fields natively used in BGV/BFV schemes. Consequently, simply porting existing linear-time PCS frameworks to the blind setting is insufficient. Table 1 provides a comparative analysis of existing code families against these critical VCoED requirements.

Table 1. Comparison of Linear Codes for PCS

Code Family	Encoding Time	Mult. Depth	Field	FHE-Friendly
Reed-Solomon[29]	$O(n \log n)$	$O(\log n)$	Non-binary	No
Brakedown[20]	$O(n)$	$O(\log n)$	Non-binary	No
EA Code[6]	$O(n \log n)$	$O(1)$	Non-binary	Yes
Binary RAA[10]	$O(n)$	$O(1)$	Binary	Yes
Ours (Non-binary RAA)	$O(n)$	$O(1)$	Non-binary	Yes

3 Preliminaries

We denote $\text{wt}(x)$ as the **Hamming Weight** of a vector $x \in \mathbb{F}_p^k$. Characters ρ, δ represent the relative hamming weights of a vector. We use capital characters such as C to represent a code and $\mathcal{C}$ as the code ensemble. For a given linear code C , we use the same character C to denote its generator matrix. The encoding process is represented as a right-multiplication of the generator matrix $c = xC$.

3.1 Coding Theory

Definition 1 (Input-Output Weight Enumerator). *For a $[n, k]$ -code C , we define the Input-Output Weight Enumerator (IOWE) as follows:*

$$A_{w,h}(C) = |\{x \in \mathbb{F}_p^k \mid \text{wt}(x) = w \cap \text{wt}(xC) = h\}|$$

Intuitively speaking, $A_{w,h}^C$ evaluates the number of codewords with weight h that are encoded from the messages weighted w .

Definition 2 (Code Ensemble and Average IOWE). *We define a **Code Ensemble** as a collection of randomized code collection $\mathcal{C} = \{C_r\}$, where r is the random seed. The **Average IOWE** of a code ensemble is defined as follows:*

$$\bar{A}_{w,h}(\mathcal{C}) = \frac{1}{|\mathcal{C}|} \sum_{C \in \mathcal{C}} A_{w,h}(C)$$

Definition 3 (Markov Process). *A **Markov process** is a stochastic process $X = \{X_t : I_N \rightarrow [0, 1]\}$ possessing the **Markov property** i.e. the following holds for all $n \in \mathbb{N}$:*

$$\Pr[X_{n+1} = x_{n+1} \mid X_n = x_n, \dots, X_1 = x_1] = \Pr[X_{n+1} = x_{n+1} \mid X_n = x_n]$$

*We can also describe the Markov property as **Memory-1**. The **transition matrix** $P^{(n)}$ of the n -th step is defined for a Markov process as follows:*

$$P_{i,j}^{(n)} = \Pr[X_{n+1} = j \mid X_n = i]$$

Definition 4 (Accumulate Code). We recall the definition of a memory-1, rate-1 convolutional linear code named **Accumulate Code** A . The generator matrix of A is defined as follows:

$$A = \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 0 & 1 & 1 & \dots & 1 \\ 0 & 0 & 1 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

We point out that the encoding behavior of A is the prefix-sum procedure, i.e. $(x_1, \dots, x_n)A = (x_1, x_1 + x_2, \dots, x_1 + \dots + x_n)$.

Definition 5 (Repeat Code). We recall the definition of a coderate- $\frac{1}{k}$ linear code named **Repeat Code** R . The generator matrix for a repeat code is defined as follows:

$$R = \begin{pmatrix} R' & & & \\ & R' & & \\ & & \ddots & \\ & & & R' \end{pmatrix} \in \mathbb{F}_p^{n \times kn}$$

where $R' = (1 \ 1 \ \dots \ 1) \in \mathbb{F}_p^k$. Intuitively speaking, the encoding procedure of R is to repeat each bit of x for k times.

3.2 (Leveled) Homomorphic Encryption

We review the definition of a leveled homomorphic encryption (LHE) scheme [9,8] that supports SIMD operations over a finite field $\mathbb{F}_p$.

Definition 6 (Leveled Homomorphic Encryption). A leveled homomorphic encryption scheme $\mathcal{E} = (\text{KeyGen}, \text{Enc}, \text{Dec}, \text{Eval})$ over a plaintext space $\mathcal{P} = \mathbb{F}_p^D$ includes the following polynomial-time algorithms:

- $\mathcal{E}.\text{KeyGen}(1^\lambda)$: Given the security parameter λ , it returns a secret key sk , a public key pk , and a public evaluation key evk .
- $\mathcal{E}.\text{Enc}_{\text{pk}}(\mathbf{m})$: Given the public key pk and a plaintext vector $\mathbf{m} \in \mathbb{F}_p^D$, it returns a ciphertext ct , also denoted as $[\mathbf{m}]$.
- $\mathcal{E}.\text{Dec}_{\text{sk}}(\text{ct})$: Given the secret key sk and a ciphertext ct , it returns the plaintext vector $\mathbf{m} \in \mathbb{F}_p^D$.
- $\mathcal{E}.\text{Eval}_{\text{evk}}(f, \{\text{ct}_i\}_{i \in [\ell]})$: Given the evaluation key evk , a function $f : \mathcal{P}^\ell \rightarrow \mathcal{P}^r$, and a set of ciphertexts, it returns a new set of ciphertexts $\{\text{ct}'_j\}_{j \in [r]}$.

The scheme is associated with a maximum multiplicative depth MAXDEPTH . For any ciphertext ct resulting from Eval , the decryption is guaranteed to be correct if and only if $\text{depth}(\text{ct}) \leq \text{MAXDEPTH}$.

Definition 7 (SIMD Packing and Encryption). Let D be the SIMD batch size (the number of available slots). The SIMD packing and encryption process consists of the following steps:

- **Plaintext Packing:** Given a vector of plaintexts $\mathbf{m} = (m_1, \dots, m_D) \in \mathbb{F}_p^D$, we define the packing operation that maps $\mathbf{m}$ into a single plaintext element $\text{pt} \in \mathcal{P}$ via the Chinese Remainder Theorem (CRT).
- **Vector Encryption:** The encryption of a packed plaintext vector $\mathbf{m}$ is denoted by $\text{ct} \leftarrow \text{Enc}_{\text{pk}}(\mathbf{m})$, which we also write in short-hand as $\llbracket \mathbf{m} \rrbracket$.

Definition 8 (Homomorphic SIMD Operations). *For any two packed ciphertexts $\llbracket \mathbf{u} \rrbracket$ and $\llbracket \mathbf{v} \rrbracket$ encrypting vectors $\mathbf{u}, \mathbf{v} \in \mathbb{F}_p^D$, and a constant plaintext vector $\mathbf{w} \in \mathbb{F}_p^D$, we define the following parallel operations:*

- **SIMD Addition:** $\llbracket \mathbf{u} \rrbracket + \llbracket \mathbf{v} \rrbracket$ computes a new ciphertext $\llbracket \mathbf{z} \rrbracket$ such that $\mathbf{z} = (u_1 + v_1, \dots, u_D + v_D)$. This operation is performed component-wise across all D slots in parallel.
- **SIMD Plaintext-Ciphertext Multiplication:** $\mathbf{w} \cdot \llbracket \mathbf{u} \rrbracket$ computes a ciphertext $\llbracket \mathbf{z} \rrbracket$ where $\mathbf{z} = (w_1 \cdot u_1, \dots, w_D \cdot u_D)$.
- **SIMD Ciphertext-Ciphertext Multiplication:** $\llbracket \mathbf{u} \rrbracket \cdot \llbracket \mathbf{v} \rrbracket$ computes a ciphertext $\llbracket \mathbf{z} \rrbracket$ where $\mathbf{z} = (u_1 \cdot v_1, \dots, u_D \cdot v_D)$.

4 Linear Minimum Distance of Generalized RAA Code

In this section, we provide a rigorous analysis of the minimum distance of Generalized RAA codes. We first review the core concepts and fundamental results and introduce the generalized RAA code. Then we analyze the distance property of the generalized RAA code with the IOWEF of the accumulator.

4.1 Warmup: Generalized RAA Codes and IOWEFs

First we recall the core concepts that are used in the analysis, including the code designs, the Input-Output Weight Enumerator Functions (IOWEFs), the composition of the codes and the spectral shapes of the RMA codes. Along with the formal definitions of these esoteric concepts, we provide intuitive comments on the definitions, aiming to offer a clearer explanation on them. Then, we list some central results on RMA codes that are carried out before [24, 6, 1, 22, 23]. The formal proofs will be presented in the appendix, since they are not the central focus of this work.

Definition 9 (Random Interleavers [24]). *For a finite set I with $|I| = n$, we can induce a rate-1 code P_π from its permutation π . More specifically, for a message $x = (x_1, \dots, x_n) \in \mathbb{F}_2^n$, $xP_\pi = (x_{\pi(1)}, \dots, x_{\pi(n)})$. We call such codes as the **Random Interleavers**.*

Definition 10 (Random Weightors). *For a random vector $\mathbf{r}$ of length n , we can induce a rate-1 code $W_{\mathbf{r}}$ from it. More specifically, for $\mathbf{r} = (r_1, \dots, r_n) \xleftarrow{\$} \mathbb{F}_p^n$, $xW_{\mathbf{r}} = \langle \mathbf{r}, x \rangle = (r_1x_1, \dots, r_nx_n)$. We call such codes as the **Random Weightors**.*

Definition 11 (Generalized Repeat-Multiple-Accumulate Code Ensemble). *For $L > 1$, we define the L -step Generalized RMA code as follows:*

$$c = xR \prod_{i=1}^L (P_{\pi_i} W_{\mathbf{r}_i} A)$$

where $\{\pi_i\}$ is a collection of random permutations on $[n]$ and $\{\mathbf{r}_i\}$ is a collection of random vectors.

Furthermore, we define the **Generalized RMA Code Ensemble** $\mathcal{C}_k$ as the collection of all such Generalized RMA codes over all possible permutation tuples $(\pi_1, \dots, \pi_L)$ and random vectors $(\mathbf{r}_1, \dots, \mathbf{r}_L)$.

Specifically, the **Generalized RAA Code Ensemble** is the Generalized RMA Code Ensemble when $L = 2$.

Now we provide some fundamental IOWE results of the related codes.

Lemma 1 (IOWE of an Accumulate code [6]). *Given block length n , the IOWE of the accumulate code A can be expressed as follows:*

$$\bar{A}_{w,h} = \sum_{i=0}^{w-1} \binom{h-1}{\lceil \frac{w-i}{2} \rceil - 1} \binom{N-h}{\lfloor \frac{w-i}{2} \rfloor} \binom{h - \lceil \frac{w-i}{2} \rceil}{i} (q-1)^{\lceil \frac{w-i}{2} \rceil} (q-2)^i$$

Lemma 2 (IOWE of a Repeat code). *Given block length n and the code rate ρ . The IOWE of the repeat code R can be expressed as follows:*

$$A_{w, \frac{w}{\rho}}(R) = \binom{\rho n}{w}$$

Lemma 3 (IOWE of composed encoders). *For two codes C_1, C_2 over field $\mathbb{F}_q$, where C_1 has block length n , the average IOWE of the code ensemble $\mathcal{C} = \{C_1 P_{\pi} W_{\mathbf{r}} C_2\}$ (i.e. the composition of C_1, C_2 via a random interleaver and a random weightor) can be expressed as the follows:*

$$\bar{A}_{w,h}(\mathcal{C}) = \sum_{l=1}^n \frac{A_{w,l}(C_1) A_{l,h}(C_2)}{\binom{n}{l} (q-1)^l}$$

Comments on lemma 3. We provide some comments on lemma 3 here. If we directly compose the two codes C_1, C_2 , the weight distribution of the final codeword will be highly related to the distributions and the values of non-zero elements in the input message, and we will derive a correlated distribution in the output. However, the random interleaver inside the composition makes the process memoryless on the distributions, while the random weightors eliminates the impact of different values. As the result, the IOWE of the composition code is a highly decoupled expression of the two IOWEs of the codes, respectively.

4.2 Linear Minimum Distance of Generalized RAA Code

In this subsection, we provide rigorous proof to the statement that Generalized RAA code has asymptotically good (i.e. linear) distance when N goes to ∞ with high probability. For the ease of presenting, we write the encoding operator $\text{RAA}(x; \pi_1, \pi_2, \mathbf{r}_1, \mathbf{r}_2) = xR(P_{\pi_1}W_{\mathbf{r}_1}A)(P_{\pi_2}W_{\mathbf{r}_2}A)$ to denote the encoding process of Generalized RAA code. When $\pi_1, \pi_2, \mathbf{r}_1, \mathbf{r}_2$ are clear in the context, we omit them and write $\text{RAA}(x)$ instead. In addition, we denote $A(w, h) = \frac{\bar{A}_{w,h}}{\binom{N}{w}(q-1)^w}$ as the normalized IOWEF of an accumulator.

More formally, our target is the following statement.

Theorem 3. *Denote S_N as the symmetric group of degree N i.e. the collection of all permutations on $\{1, 2, \dots, N\}$. Denote k as the repeat times of the repeat code.*

There exists a constant $K \in \mathbb{N}^+$ such that for any $k > K$, the following holds:

$$\Pr_{\substack{\pi_1, \pi_2 \xleftarrow{\$} S_N \\ \mathbf{r}_1, \mathbf{r}_2 \xleftarrow{\$} \mathbb{F}_p^N}} [\exists x \text{ s.t. } \text{wt}(\text{RAA}(x)) \leq \delta N] = o\left(\frac{1}{N}\right)$$

Proof Sketch of Theorem 3 To prove Theorem 3, we employ a union bound analysis over the space of all possible non-zero inputs x . Specifically, we transform the original probability taken over the random choice of permutations π_1, π_2 and weighting vectors $\mathbf{r}_1, \mathbf{r}_2$ into a summation of probabilities that the weight of the output codeword $\text{wt}(\text{RAA}(x))$ is bounded by δN for a fixed input x .

However, applying a naive union bound over all q^N possible inputs $x \in \mathbb{F}_p^N \setminus \{0\}$ results in an excessively loose estimation. For instance, consider the case where $\text{wt}(x) = 1$. The probability $\Pr[\text{wt}(\text{RAA}(x)) \leq \delta N]$ degrades to a polynomially small factor, specifically $O(N^{-k})$. Since the total number of inputs grows exponentially as q^N , a uniform bound would fail to show the probability is $o(1)$. Therefore, we must partition the input space by weight and apply a stratified union bound, treating inputs of different weights with separate analytical techniques.

$$\Pr_{\pi_1, \pi_2 \xleftarrow{\$} S_N} [\exists x \text{ s.t. } \text{wt}(\text{RAA}(x)) \leq \delta N] \quad (1)$$

$$\Pr_{\pi_1, \pi_2 \xleftarrow{\$} S_N, \mathbf{r}_1, \mathbf{r}_2 \xleftarrow{\$} \mathbb{F}_p^N} [\cup_x \text{wt}(\text{RAA}(x)) \leq \delta N] \quad (2)$$

$$\Pr_{\pi_1, \pi_2 \xleftarrow{\$} S_N, \mathbf{r}_1, \mathbf{r}_2 \xleftarrow{\$} \mathbb{F}_p^N} [\cup_w (\cup_{\text{wt}(x)=w} \text{wt}(\text{RAA}(x)) \leq \delta N)] \quad (3)$$

$$\leq \sum_{w=1}^N \Pr_{\pi_1, \pi_2 \xleftarrow{\$} S_N, \mathbf{r}_1, \mathbf{r}_2 \xleftarrow{\$} \mathbb{F}_p^N} [\cup_x \text{wt}(\text{RAA}(x)) \leq \delta N | \text{wt}(x) = w] \quad (4)$$

Now for a fixed input weight w , the space of all input x is of size $\binom{N}{w}(q-1)^w$. Then, for an arbitrary input x of weight w , we are now analyzing the weight distribution evolution during the encoding process:

1. **Repeater:** Given the input x of weight w , the weight of the output codeword after the repeater must be kw .
2. **Random Interleaver + Random Weightor:** Since both operators do not bring any influence to the weight of the codeword, the weight of codeword is still kw after these two operators.
3. **Accumulator:** Now the weight distribution of $xR P_{\pi_1} W_{\mathbf{r}_1} A$ after the accumulator is exactly the (kw) -th row of the IOWE matrix of an accumulator of degree N i.e.

$$\Pr_{\pi_1, \mathbf{r}_1} [\text{wt}(xR(P_{\pi_1} W_{\mathbf{r}_1} A)) = s] = A(kw, s)$$

4. **Second Random Interleaver + Random Weightor + Accumulator:** Similarly to the analysis of the first interleaver and weightor, these two operators do not influence the weight of the codeword. Therefore, we can focus on the question that how it performs when a s -weight input is fed into the second accumulator. Using the IOWEF, we can answer it immediately

$$\Pr_{\pi_2, \mathbf{r}_2} [\text{wt}(x P_{\pi_2} W_{\mathbf{r}_2} A) = h | \text{wt}(x) = s] = A(s, h)$$

Putting all together, we have derived the following expression for the probability of obtaining an h -weight output codeword given a w -weight input x .

$$\Pr_{\pi_1, \pi_2, \mathbf{r}_1, \mathbf{r}_2} [\text{wt}(\text{RAA}(x)) = \text{wt}(xR(P_{\pi_1} W_{\mathbf{r}_1} A)(P_{\pi_2} W_{\mathbf{r}_2} A)) = h] = \sum_{s=1}^N A(kw, s) \cdot A(s, h)$$

Now since we are only interested in the case when the output codeword is of weight no more than δN , we write $D(s) = \sum_{h=1}^{\delta N} A(s, h)$ and the above probability

becomes $\sum_{s=1}^N A(kw, s) \cdot D(s)$. Therefore, we can take the union bound on the equation 4:

$$\Pr_{\substack{\pi_1, \pi_2 \xleftarrow{\$} S_N \\ \mathbf{r}_1, \mathbf{r}_2 \xleftarrow{\$} \mathbb{F}_p^N}} [\cup_x \text{wt}(\text{RAA}(x)) \leq \delta N | \text{wt}(x) = w] \leq \left(\frac{N}{k}\right) (q-1)^w \sum_{s=1}^N A(kw, s) \cdot D(s) \quad (5)$$

Our objective is to demonstrate that, by taking a union bound over all possible inputs for any given input weight w , the probability that the final output weight is less than or equal to δN is $o(\frac{1}{N^2})$, which is formally presented in Theorem 4. Note that if Theorem 4 is correct, we can take the summation of equation 5 to complete the proof of Theorem 3:

$$\sum_{w=1}^N \Pr_{\substack{\pi_1, \pi_2 \xleftarrow{\$} S_N \\ \mathbf{r}_1, \mathbf{r}_2 \xleftarrow{\$} \mathbb{F}_p^N}} [\cup_x \text{wt}(\text{RAA}(x)) \leq \delta N | \text{wt}(x) = w] \leq \sum_{w=1}^N o(\frac{1}{N^2}) = o(\frac{1}{N})$$

Theorem 4 (Central Theorem). *There exists $K \in \mathbb{N}^+$ such that for any $k > K$, we have the following:*

$$\left(\frac{N}{k}\right) (q-1)^w \sum_{s=1}^N A(kw, s) \cdot D(s) = o(\frac{1}{N^2})$$

Remark The probabilistic suppression characterized by Theorem 4 involves intricate algebraic transitions across different weight regimes. To provide a more transparent illustration of the underlying operational intuition, we refer the readers to the numerical analysis in Section 5.

Proof Sketch of Theorem 4 To establish the soundness of our construction, we provide a two-fold proof sketch for Theorem 4 (Central Theorem). This theorem demonstrates that for any given input weight w , the probability of the final output weight falling below the threshold δN is polynomially small $o(1/N^2)$.

The analysis is partitioned based on the magnitude of the input weight w relative to the block size N :

1. **The First Fold: Case for Large Input Weight** ($w \geq \frac{0.01N}{k}$): When the input weight is large, the union bound term becomes exponentially small due to the concentration properties of the Accumulator kernel.
 - **Kernel Sharpness:** A critical observation is that for large w , the w -th row of the Accumulator kernel $A(w, h)$ is extremely "sharp." This concentration implies that the mass of the distribution is pushed far away from the low-weight region.

- **Negligible Cumulative Distribution:** Because the distribution is concentrated at higher weights, the sum of the first δN elements becomes negligible.
 - **Exponential Decay:** As formalized in Theorem 5 and Corollary 1, this leads to an upper bound of the form $o(e^{-CN})$ for some constant $C > 0$, ensuring the probability of a low-weight output is suppressed exponentially.
2. **The Second Fold: Case for Small Input Weight ($w < \frac{0.01N}{k}$):** The analysis of relatively small input weights constitutes the most intricate portion of the proof, as the standard asymptotic tools lose their precision in this regime.
- **Failure of Stirling's Formula:** A fundamental analytical challenge arises when the input weight w is extremely small. Specifically, in the case of $w = 1$, the union bound term degenerates from an exponential form to one of polynomial magnitude. Since Stirling's formula is intrinsically designed to handle terms with exponential growth, it becomes effectively inapplicable.
 - **Analytical Induction:** To tackle this problem, we rely on a tighter estimation based on fundamental inequalities rather than raw asymptotics. This approach lacks a simple intuitive shortcut and instead requires comprehensive analytical inductions across the weight spectrum.
 - **Bounding via Lemmas 4 and 5:** By establishing specific bounds for $A(w, h)$ and $D(w)$ in the small-weight regime (as seen in Lemma 3), we successfully derive a cumulative upper bound of $O(1/N^2)$ for the sum. This ensures that even for very low-weight inputs, the probability of the system failing to expand the weight remains within acceptable limits for sound security.

Theorem 5 (Estimation of $D(w)$ for large w). *When $\delta < 0.2$, there exists a constant $\Lambda > 0$ such that the following holds:*

$$D(w) \leq e^{-\Lambda w}$$

for all $w \geq 0.01N$.

Proof Sketch of Theorem 5

1. **Asymptotic Estimation via Stirling's Approximation:** Our first objective is to provide a precise estimation of the sum $\sum_{w \in L} K_N(i, w)$. By applying Stirling's Formula $\frac{1}{n+1}e^{nH(\frac{k}{n})} \leq \binom{n}{k} \leq e^{nH(\frac{k}{n})}$ to the combinatorial terms inherent in the Markov kernel K_N , we prove that, up to a polynomial factor $\text{poly}(N)$, the summation adheres to the following exponential pattern:

$$\sum_{w \in L} K_N(i, w) = O(\text{poly}(N) \cdot e^{\Phi(\cdot)N})$$

where $\Phi(\cdot)$ is a function derived from the entropy-like terms of the weight distribution.

2. **Rigorous Analysis of the Exponent $\Phi(\cdot)$:** We conduct an exhaustive analysis of its functional properties, including its monotonicity and evaluations at critical boundary points within the relevant domain. Our analysis strictly demonstrates that $\Phi(\cdot)$ remains strictly negative across the entire range of interest. Specifically, we prove that: $\Phi(\cdot) < -c$ for some positive constant $c > 0$.

Corollary 1. *There exists $K \in \mathbb{N}^+$ and a $C(K, q) > 0$ such that for any $k > K, w \geq \frac{0.01N}{k}$, we have the following:*

$$\binom{\frac{N}{k}}{w} (q-1)^w D(kw) = o(e^{-CN})$$

Corollary 2. *There exists $K \in \mathbb{N}^+$ such that for any $k > K, w \leq \frac{0.01N}{k}$, we have the following:*

$$\binom{\frac{N}{k}}{w} (q-1)^w \sum_{s=0.01N}^{\delta N} D(s) = o(e^{-CN})$$

Unlike the large-weight regime, the union bound for $w = 1$ collapses into a polynomial magnitude, a domain where Stirling's formula loses its predictive power. This necessitates a distinct analytical strategy: instead of relying on exponential suppression, we develop an alternative estimation framework that leverages the granular properties of the Accumulator kernel to ensure soundness.

Lemma 4. *For $w \leq 0.01N$, we have the following estimation for $A(w, h)$:*

$$A(w, h) \leq \frac{w^2}{N} \cdot \left(\frac{h}{N-w}\right)^{\frac{w}{2}} \cdot 3^w$$

Furthermore, we have the following estimation for $D(w)$:

$$D(w) \leq \frac{2w^2}{w+1} \cdot \frac{\delta N + 1}{N} \cdot \left(\frac{\delta N + 1}{N-w}\right)^{\frac{w}{2}} \cdot 3^w$$

Proof Insight of Lemma 4 The complexity of the Accumulator kernel $A(w, h)$ necessitates a departure from standard asymptotic approximations in favor of a granular, factor-by-factor estimation. Our approach is guided by three primary analytical insights:

1. **One-to-One Factor Assignment:** We treat the binomial coefficient $\binom{N}{w}$ as the product of w large integers divided by $w!$. By factoring out these factorial components, we ensure that the number of multipliers in the numerator of the kernel expression exactly matches the number of multipliers in the denominator. This alignment allows us to perform a one-to-one assignment of terms, facilitating a controlled estimation of the entire product.

2. **Strategic Inequality Application:** To simplify the resulting product of terms like $(h-1) \cdots (h-i)$ and $(N-h) \cdots (N-h-j+1)$, we apply a fundamental inequality. Specifically, we utilize the observation that the function $(h-x)(N-h-j+x)$ is decreasing for $x \geq 0$. This allows us to uniformly scale the pairs in the numerator as follows:

$$(h-x)(N-h-j+x) \leq h(N-h-j)$$

These scaled terms are then assigned to their corresponding counterparts in the denominator to keep the bound analytically tractable.

3. **Regime-Specific Simplification:** Our analysis focuses on the regime where the weight and height are significantly smaller than the block size ($w \ll N, h \ll N$). In this context, we apply two key simplifications to the denominator:

- The denominator terms are uniformly bounded by $(N-w)$.
- The ratio $\frac{N-h-a+1}{N-w}$ is conservatively scaled to 1.

Lemma 5. *There exists $K \in \mathbb{N}^+$ such that for any $k > K, w \leq \frac{0.01N}{k}$, we have the following:*

$$\left(\frac{N}{k}\right) (q-1)^w \sum_{s=1}^{0.01N} A(kw, s) \cdot D(s) = O\left(\frac{1}{N^2}\right)$$

Proof Sketch of Lemma 5 The proof begins by substituting the fine-grained upper bounds for $A(kw, s)$ and $D(s)$ derived in Lemma 2. We perform a pure algebraic reduction on the product $A(kw, s) \cdot D(s)$, treating it as a function of the intermediate weight s within the summation interval $[1, 0.01N]$. Through a first-order derivative analysis, we analyze the monotonicity of the resulting expression across the weight spectrum.

A key analytical finding of our induction is that the function $A(kw, s) \cdot D(s)$ does not show a single monotonicity within the interval $[1, 0.01N]$. However, the function is convex, implying that its maximum value must be attained at one of the two endpoints of the interval: $s = 1$ or $s = 0.01N$.

To finalize the proof, we analyze the evaluations at these two critical boundaries and derive a complete proof for the statement.

Theorem 6 (Estimation of small w). *There exists $K \in \mathbb{N}^+$ such that for any $k > K, w \leq \frac{0.01N}{k}$, we have the following:*

$$\left(\frac{N}{k}\right) (q-1)^w \sum_{s=1}^N A(kw, s) \cdot D(s) = o\left(\frac{1}{N^2}\right)$$

Proof. This is an immediate result by combining Corollary 2 and Lemma 5. $\square$

We end up this section by emphasizing the following properties of the Generalized RAA code without proof. The properties are crucial in the following sections.

Theorem 7. *The Generalized RAA Codes have the following properties:*

- **Constant Multiplication Depth:** *RAA Codes have exactly $k = O(1)$ multiplication depth, where k is the iteration of Accumulators.*
- **Linear-Time Encodable:** *The computation of encoding a message of length m costs $O(m)$ additions and $O(m)$ multiplications.*
- **Compatible with Arbitrary Fields:** *RAA Codes have a good distance with a constant rate in an arbitrary field.*

5 Numerical Analysis

In Section 4, we established a rigorous analytical framework characterizing the minimum distance property of generalized RAA codes. This derivation necessitated a series of intricate maneuvers within the algebraic domain, specifically regarding the monotonicity and extremum analysis of complicated expressions. While these proofs provide the foundational certainty required for cryptographic protocols, the sheer density of the symbolic manipulation can be inherently "opaque" to the reader.

We acknowledge that a purely formalistic approach, though logically complete, may fail to convey the underlying operational intuition that governs the system's evolution. Without a clear conceptual background, the transition between these algebraic states can appear as a "black box", masking the core insights that drive our analysis.

Consequently, this section is dedicated to a numerical investigation of the pivotal expressions identified in our theoretical analysis. By grounding our abstract results in visual representations, we aim to provide a more transparent illustration.

Specifically, we conduct our numerical analysis using a representative parameter set where the block length $N = 2^{20}$, the field size $q = 2^{16} + 1$, and the relative distance threshold $\delta = 0.25$. This configuration is purposefully selected to align with the concrete evaluation parameters established in Section 7. By utilizing the same benchmarks for both our theoretical visualization and empirical performance evaluation, we ensure consistency across the paper and demonstrate the practical applicability of our analytical bounds.

We list the central expressions that will be illustrated in this section below:

$$A(h; w, q) = \frac{\sum_{i=0}^{w-1} \binom{h-1}{\lceil \frac{w-i}{2} \rceil - 1} \binom{N-h}{\lfloor \frac{w-i}{2} \rfloor} \binom{h - \lceil \frac{w-i}{2} \rceil}{i} (q-1)^{\lceil \frac{w-i}{2} \rceil} (q-2)^i}{\binom{N}{w} (q-1)^w}$$

$$D(w; \delta, q) = \sum_{h=1}^{\delta N} A(h; w, q)$$

$$\tilde{A}(h; w, k, q) = \binom{\frac{N}{k}}{w} (q-1)^w A(h; kw, q)$$

$$P_{k, \delta, q}(h|w) = \tilde{A}(h; w, k, q) \cdot D(h; \delta, q)$$

And we emphasize that our target union bound probability expression is

$$\begin{aligned}
\Pr_{C \leftarrow \mathcal{C}} [\cup_x \text{wt}(xC) < \delta N] &\leq \sum_{w=1}^N \Pr_{C \leftarrow \mathcal{C}} [\cup_x \text{wt}(xC) < \delta N | \text{wt}(x) = w] \\
&\leq \sum_{w=1}^N \binom{\frac{N}{k}}{w} (q-1)^w \sum_{h=1}^N A(h; kw, q) \cdot D(w; \delta, q) \\
&= \sum_{w=1}^N \sum_{h=1}^N P_{k, \delta, q}(h|w)
\end{aligned}$$

5.1 Numerical Behavior of the Accumulator IOWE $A(h; w, q)$

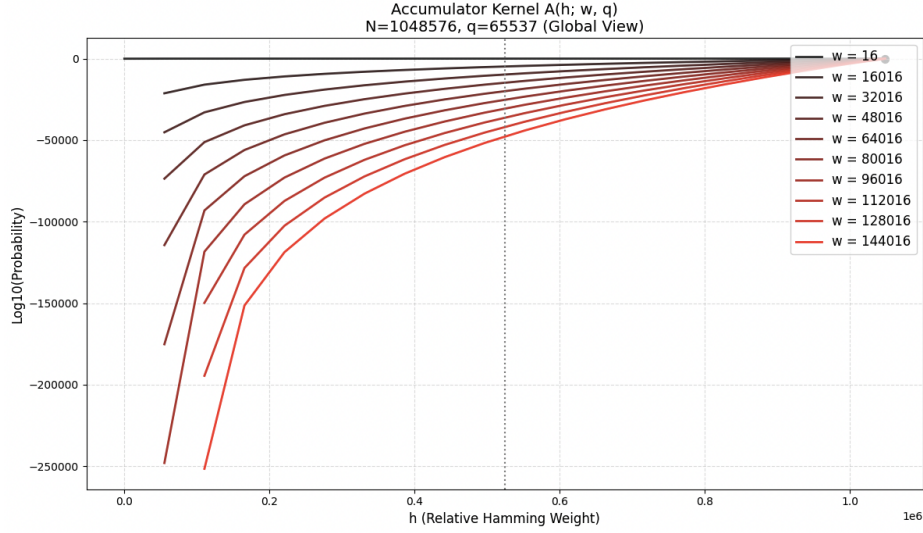


Fig. 1. Numerical evaluation of $\log A(h; w, q)$ for $N = 2^{20}$ and $q = 2^{16} + 1$. The curves illustrate the evolution of the output relative Hamming weight h across various input weights w .

A closer numerical inspection of the $A(h; w, q)$ curves in Figure 1 reveals a critical phase transition in the concentration behavior of the output weight h . While the algebraic complexity of the combinatorial sum initially masks this, our concrete analysis identifies two distinct regimes determined by the input weight w :

1. **The Exponential Decay Regime** ($w > 0.01N$): When the input weight w is not too small, specifically $w > 0.01N$, the function exhibits an exponentially small probability mass in the low-Hamming-weight regions (where

$h \ll (1 - \frac{1}{q})N$). As seen in the steeper red curves, the log-probability drops rapidly as h moves toward zero. To contextualize this phenomenon, we emphasize that in the such case the behavior is primarily governed by the first and third binomial terms in the numerator $\binom{h-1}{\lceil \frac{w-i}{2} \rceil - 1}$ and $\binom{h - \lfloor \frac{w-i}{2} \rfloor}{i}$. As h decreases, these terms become the decisive factors. Crucially, these terms are subject to an exponent roughly on the order of $w/2$. In the linear regime where w is a constant fraction of N , the Stirling approximation reveals that the product of these terms scales as $(h/N)^{O(w)}$. Consequently, even a relatively "small" weight w (such as $0.01N$) results in a high-degree power that suppresses the probability mass to exponentially small levels in the low- h region.

2. **The Polynomial Degeneration Regime (Small w):** Conversely, for very small input weights (where w is sub-linear or even constant), we observe a notable "flattening" of the curves. In this regime, the superior exponential concentration properties degrade into a polynomial-level smallness. The dark curves in Figure 1 illustrate this transition: the gradient is significantly shallower at low h values. When w is sub-linear, the $w/2$ "exponent" in $\binom{h-1}{\lceil \frac{w-i}{2} \rceil - 1}$ and $\binom{h - \lfloor \frac{w-i}{2} \rfloor}{i}$ terms is too weak to trigger rapid suppression, causing the function to degenerate into the polynomial regime described previously.

5.2 Numerical Behavior of the Cumulative Kernel $D(w; \delta, q)$

Building upon our understanding of the point-wise transition $A(h; w, q)$, we now examine the cumulative distribution function $D(w)$. This expression captures the aggregate probability that a codeword of weight w fails to achieve a minimum distance threshold δ after the second accumulation. We emphasize the following key illustration from Figure 2.

1. **Linear Logarithmic Decay and Stability:** The most prominent feature in Figure 2 is that the log-probability decreases almost linearly as the input weight w increases. This suggests that the total failure probability $D(w)$ suppresses at a consistent exponential rate relative to w . When w is extremely small (near the origin), the function approaches 0 on the log scale. This represents the regime where sparse inputs lack the combinatorial entropy to guarantee expansion. As soon as w begins to scale, the function becomes strictly decreasing and significantly negative. For $w = 10^5$, the probability reaches a negligible $10^{-19751.3}$, demonstrating that the leakage of codewords into the low-weight region is suppressed to a degree that far exceeds standard cryptographic requirements.

The remarkable linear decay observed in the cumulative function $D(w)$ is not an independent phenomenon. It is a direct structural consequence of the point-wise behavior of $A(h; w, q)$ analyzed in the previous subsection. By observing the "velocity" at which the curves in Figure 1 shift, the reader can intuitively predict the trajectory of Figure 2.

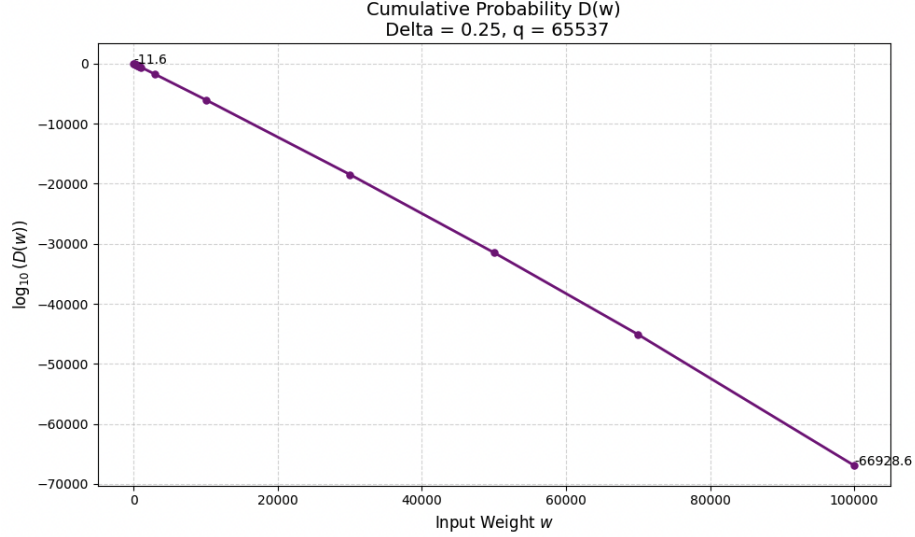


Fig. 2. Logarithmic scaling of the cumulative failure probability $D(w; \delta, q)$ as a function of input weight w for $\delta = 0.25$ and $q = 65537$. The near-linear trajectory indicates a consistent exponential suppression of low-weight outputs, while the subtle downward curvature (negative second derivative) suggests an accelerated convergence for denser input messages.

Regarding the above concrete analysis on $A(h; w, q)$ and $D(w; \delta, q)$, we can figure it out that when w is not too small (corresponding the "Exponential Decay Regime" paragraph of $A(h; w, q)$), $D(w; \delta, q)$ is exponentially small.

5.3 Numerical Behavior of the Union Bound Probability $P_{k,\delta,q}(h|w)$

Finally, we examine the behavior of the composite function $P_{k,\delta,q}(h|w) = \tilde{A}(h; w, k, q) \cdot D(h; \delta, q)$, as visualized in Figure 3. This function represents the probability of a specific output weight h occurring given an input weight w and a target distance threshold δ . The numerical results reveal a crucial bifurcation in the error probability behavior, which directly dictates our subsequent proof strategy. We illustrate this figure from the following two aspects:

1. **The Sparse Regime: Polynomial Degeneration.** For extremely small input weights (illustrated by the curves for $w = 8, 16, 32, 64$), the function behaves in a qualitatively different manner compared to the linear regime. In this regime, the curves are strictly decreasing. There is no mixing peak, instead, the highest probability of error occurs at the minimum possible output weight. Critically, at the origin point ($h \approx 0$), the log-probability is near zero. This indicates that for sparse inputs, the probability of a low-weight output is only polynomially small rather than exponentially small.

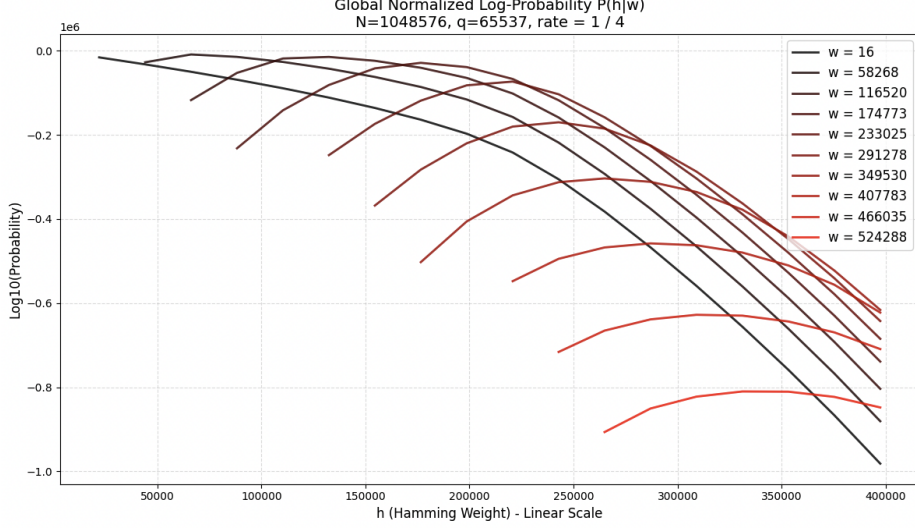


Fig. 3. Numerical distribution of the conditional probability $P(h|w)$ for the Generalized RAA code. The curves are plotted with parameters $N = 2^{20}$, $q = 2^{16} + 1$, and a relative distance threshold of $\delta = 0.25$. The result empirically validates the union bound analysis by demonstrating the exponential decay of the probability that a codeword has low Hamming weight h across varying input weights w .

While these events are still rare, they do not benefit from the aggressive exponential suppression seen in denser codewords.

2. **The Linear Regime: Emergence of the Concentration Peak.** As w increases into the linear regime (e.g., $w \geq 58268$), the function undergoes a phase transition. The curves initially increase to a maximum value before decaying. Significantly, as w continues to grow, the global maximum of each curve (the peak) shifts downward. Even at its most likely point, the probability remains strictly and significantly negative on the log scale, indicating that error events become exponentially unlikely.

The numerical evidence in Figure 3 highlights that the sparse regime ($w \leq 16$) represents the most dangerous region for our proof. In this area, the combinatorial entropy is insufficient to trigger exponential decay, and the proof relies on polynomial bounds. This observation serves as the primary inspiration for our proof architecture in Section 4:

1. Lemma 5 and Theorem 6 (Sparse Cases): We isolate the analysis for small w into dedicated theorems. Recognizing that exponential bounds are unattainable here, we weaken the claim to polynomial smallness, specifically proving that the failure probability is $o(\frac{1}{N^2})$.
2. Theorem 5 and Corollary 1 (Linear Case): For the remaining weight distribution, we leverage the exponential suppression demonstrated by the downward-shifting peaks in the linear regime to provide the proofs.

5.4 Concrete Numbers

To conclude, we provide in Table 2 achievable parameters for Generalized RAA codes, along with provable failure bounds.

Table 2. Concrete Failure Probability Bounds $\log_{10}(P(h|w))$ for Generalized RAA Codes. The data demonstrates the relationship between the circuit size N and the code rate ρ .

Rate ρ / Size N	2^{18}	2^{20}	2^{22}	2^{24}	2^{26}	2^{28}	2^{30}
1/4	-8.099	-7.990	-7.830	-7.650	-7.470	-7.290	-7.110
1/8	-27.210	-28.580	-29.755	-30.820	-31.855	-32.879	-33.902

As demonstrated in Table 2, the failure probability is highly sensitive to the choice of the code rate ρ . For the scale interval between 2^{20} and 2^{30} , we can adopt aggressive parameters ($\rho = 1/4$) to achieve a faster prover. In this regime, the failure probability remains strictly bounded below $10^{-7.11}$, providing a practical balance between performance and security. However, for larger-scale circuits beyond this range, we recommend selecting more conservative parameters such as $\rho = 1/8$. Under this configuration, the soundness improves significantly with the growth of N , achieving a failure probability of $10^{-33.90}$ at $N = 2^{30}$. This ensures robust security guarantees even as the circuit complexity increases, effectively mitigating the slight upward trend observed in the more aggressive configuration.

6 Linear-time Blind PCS With Generalized RAA Code

In this section, we present the construction and optimization of our linear-time blind PCS.

First, we formally define the concept of blind PCS. Then, we construct the blind PCS by building upon the framework proposed in [2] and further developed in [20,30], adapting their methodology to support encrypted polynomial commitments natively. Building on this foundation, we incorporate the SIMD packing techniques pioneered by *Phalanx* to achieve concrete acceleration in homomorphic operations. This optimization allows our prover to exploit the inherent parallelism of BGV/BFV schemes, significantly reducing the computational overhead of plaintext-ciphertext multiplications.

Finally, we provide a discussion on the choice of the underlying field $\mathbb{F}_p$ to ensure native compatibility with FHE parameters, followed by a rigorous security analysis of the full protocol. We demonstrate that our scheme achieves strictly linear prover complexity while maintaining the robust soundness guarantees required for VCoED.

6.1 Notations

We use B as the SIMD batch size of the underlying FHE scheme. We use $\llbracket m \rrbracket$ to represent a ciphertext containing the plaintext vector $m = (m_1, \dots, m_B)$. We typically use the notation $C : \mathbb{F}^{\rho N} \rightarrow \mathbb{F}^N$ to denote a general linear code with a constant code rate $0 < \rho < 1$ and with the codeword length N (and the message length is ρN).

6.2 Blind Polynomial Commitment Scheme

In this section, we introduce the Blind Polynomial Commitment Scheme (Blind PCS), which enables a prover to commit to a polynomial and prove its evaluations while the coefficients remain encrypted under a leveled homomorphic encryption (HE) scheme.

Definition 12 (Blind PCS). *For a given leveled HE scheme $\mathcal{E} = (\text{KeyGen}, \text{Enc}, \text{Dec}, \text{Eval})$ and a security parameter λ , a Blind PCS consists of a tuple of algorithms $(\text{Setup}, \text{Commit}, \text{OpenPoly}, \text{Eval})$ defined over the encrypted domain:*

- $\text{Setup}(1^\lambda, \mu) \rightarrow \text{pp}$: Takes security parameter λ and number of variables μ , outputs public parameters pp .
- $\text{Commit}(\text{pp}, \llbracket \tilde{f} \rrbracket) \rightarrow (C, \mathcal{D})$: Takes a ciphertext vector $\llbracket \tilde{f} \rrbracket$ encrypting the coefficients of a μ -variate multilinear polynomial $\tilde{f}$. It outputs a commitment C and auxiliary message $\mathcal{D}$.
- $\text{Commit}(\text{pp}, \llbracket \tilde{f} \rrbracket) \rightarrow (C, \mathcal{D})$: Takes a ciphertext vector $\llbracket \tilde{f} \rrbracket$ encrypting the coefficients of a μ -variate multilinear polynomial $\tilde{f}$. It outputs a commitment C and auxiliary message $\mathcal{D}$.
- $\text{OpenPoly}(\text{pp}, C, \llbracket \tilde{f} \rrbracket, \mathcal{D}) \rightarrow b$: Verifies the commitment given the encrypted polynomial $\llbracket \tilde{f} \rrbracket$, outputting a bit $b \in \{0, 1\}$.
- $\text{Eval}(\text{pp}, C, z, \llbracket y \rrbracket, \mu, \llbracket \tilde{f} \rrbracket, \mathcal{D}) \rightarrow b$: An interactive Argument of Knowledge (AoK) between a Prover $\mathcal{P}$ and Verifier $\mathcal{V}$. $\mathcal{P}$ proves that the underlying plaintext satisfies $\tilde{f}(z) = y$, where $\llbracket y \rrbracket$ is an encrypted evaluation result. The algorithm outputs $b = 1$ if the verification passes.

A Blind PCS must satisfy the following properties within the noise budget of the HE scheme:

- **Completeness**: For all $\lambda, \mu \in \mathbb{N}$, $(\text{pk}, \text{sk}) \leftarrow \mathcal{E}.\text{KeyGen}(1^\lambda)$, any polynomial $\tilde{f}$ and point z , let $\llbracket \tilde{f} \rrbracket \leftarrow \mathcal{E}.\text{Enc}_{\text{pk}}(\tilde{f})$ and $\llbracket y \rrbracket \leftarrow \mathcal{E}.\text{Enc}_{\text{pk}}(\tilde{f}(z))$. The following probability equals 1:

$$\Pr \left[\text{Eval}(\text{pp}, C, z, \llbracket y \rrbracket, \mu, \llbracket \tilde{f} \rrbracket, \mathcal{D}) = 1 \mid \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda, \mu) \\ (C, \mathcal{D}) \leftarrow \text{Commit}(\text{pp}, \llbracket \tilde{f} \rrbracket) \end{array} \right] = 1$$

- **Binding:** For any PPT adversary $\mathcal{A}$, the probability that $\mathcal{A}$ can produce a commitment C that can be validly opened to two different encrypted polynomials $\llbracket \tilde{f}_0 \rrbracket$ and $\llbracket \tilde{f}_1 \rrbracket$ (where $\mathcal{E}.\text{Dec}_{\text{sk}}(\llbracket \tilde{f}_0 \rrbracket) \neq \mathcal{E}.\text{Dec}_{\text{sk}}(\llbracket \tilde{f}_1 \rrbracket)$) is negligible:

$$\Pr \left[\begin{array}{c} \text{Dec}(\text{sk}, \llbracket \tilde{f}_0 \rrbracket) \neq \text{Dec}(\text{sk}, \llbracket \tilde{f}_1 \rrbracket) \\ \wedge \text{OpenPoly}(\text{pp}, C, \llbracket \tilde{f}_0 \rrbracket, \mathcal{D}_0) = 1 \\ \wedge \text{OpenPoly}(\text{pp}, C, \llbracket \tilde{f}_1 \rrbracket, \mathcal{D}_1) = 1 \end{array} \middle| \begin{array}{c} \text{pp} \leftarrow \text{Setup}(1^\lambda, \mu) \\ (C, \llbracket \tilde{f}_0 \rrbracket, \mathcal{D}_0, \llbracket \tilde{f}_1 \rrbracket, \mathcal{D}_1) \\ \leftarrow \mathcal{A}(\text{pp}, \text{pk}) \end{array} \right] \leq \text{negl}(\lambda)$$

- **Knowledge Soundness:** There exists a PPT oracle machine $\mathcal{Ext}$ such that for any PPT adversary $(\mathcal{A}_1, \mathcal{A}_2)$, if $\mathcal{A}_2$ convinces the verifier that $\tilde{f}(z) = y$ over ciphertexts, then $\mathcal{Ext}$ can extract a valid polynomial $\tilde{f}$ and auxiliary message $\mathcal{D}$ from the adversary's state such that the plaintext relation holds:

$$\Pr \left[\begin{array}{c} \langle \mathcal{A}_2(st), \mathcal{V} \rangle(\text{pp}, C, z, \llbracket y \rrbracket) = 1 \\ \wedge (\text{Dec}(\text{sk}, \llbracket \tilde{f} \rrbracket)(z) \neq \text{Dec}(\text{sk}, \llbracket y \rrbracket)) \end{array} \middle| \begin{array}{c} (C, z, \llbracket y \rrbracket, st) \leftarrow \mathcal{A}_1(\text{pp}, \text{pk}) \\ (\tilde{f}, \mathcal{D}) \leftarrow \mathcal{Ext}^{\mathcal{A}_1, \mathcal{A}_2} \end{array} \right] \leq \text{negl}(\lambda)$$

6.3 Protocol

The foundation of our blind construction is the Tensor Code framework proposed in Ligerio [2], extended to operate over encrypted witness.

SIMD Packing To reduce the computational latency and communication overhead of our protocol, we consider to leverage the SIMD capabilities of FHE schemes. This optimization allows us to perform batch encoding and proving on encrypted polynomials with significantly improved throughput.

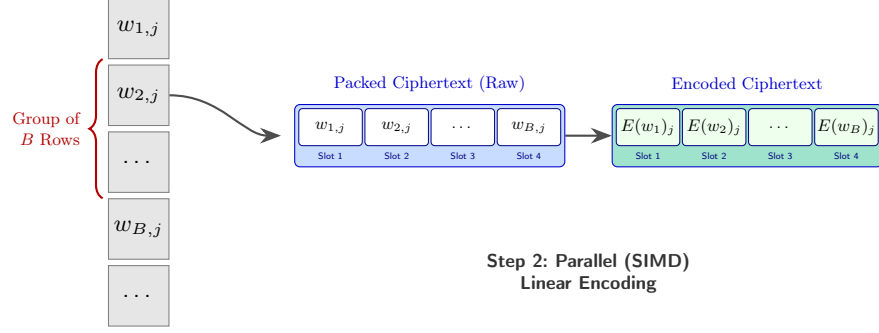
To optimize computational throughput, we follow the packing strategy proposed in *Phalanx*[32], where B consecutive rows from each column of the coefficient matrix are packed into a single ciphertext. This structure enables the system to leverage FHE SIMD capabilities to perform homomorphic encoding and random linear combinations in parallel across all B rows. The detailed packing process is illustrated in Figure 4.

Theorem 8. *There exists a blind Polynomial Commitment Scheme with:*

- **Prover Complexity:** $O(N)$ plaintext-ciphertext multiplications.
- **Verifier Complexity:** $O(\lambda\sqrt{N})$ plaintext-plaintext multiplications and $O(\lambda\sqrt{N})$ decryptions.
- **Proof Size:** $O(\lambda\sqrt{N})$ ciphertexts and $O(\lambda \log N)$ hashes.
- **Multiplication depth on ciphertexts:** $O(1)$ multiplication depth.

Protocol. For a polynomial $f(x) = \sum_{i=1}^n f_i x^i$, we recall the following protocol that enables the prover to prove the evaluation of any point of the polynomial with $O(n)$ time complexity. The prover commits to an $r \times m$ matrix of encrypted coefficients. Consider a linear code $C : \mathbb{F}^m \rightarrow \mathbb{F}^M$, the prover and the verifier interact corresponding to the following steps:

Fig. 4. The SIMD-based ciphertext packing and aggregation process in our Blind PCS.
Witness Matrix (Column j)



Step 1: Matrix Arrangement
& Batch Packing

Blind Polynomial Commitment Scheme

- **Commit Phase.**

1. $\mathcal{P}$ encodes each row of the encrypted matrix:
 $\llbracket c_i \rrbracket = \text{Enc}_C(\llbracket f_{i,1} \rrbracket, \llbracket f_{i,2} \rrbracket, \dots, \llbracket f_{i,m} \rrbracket)$ and derives an encoded and encrypted matrix.
2. $\mathcal{P}$ computes the Merkle root of the nodes $\llbracket c_1 \rrbracket, \llbracket c_2 \rrbracket, \dots, \llbracket c_r \rrbracket$ i.e.
 $\text{com}_{MT} = \text{MerkleTree.Commit}(\llbracket c_1 \rrbracket, \llbracket c_2 \rrbracket, \dots, \llbracket c_r \rrbracket)$ and sends the root hash com_{MT} to $\mathcal{V}$.

- **Testing Phase.**

1. $\mathcal{V}$ samples a random plain vector $\gamma \xleftarrow{\$} \mathcal{F}_p^r$ and sends to $\mathcal{P}$.
2. $\mathcal{P}$ computes the plaintext-ciphertext multiplication
 $\llbracket f' \rrbracket \leftarrow \langle \gamma, (\llbracket f_1 \rrbracket, \dots, \llbracket f_r \rrbracket) \rangle$ and sends $\llbracket f' \rrbracket$ to $\mathcal{V}$.
3. $\mathcal{V}$ decrypts and reads the random linear combination $\llbracket f' \rrbracket$ as a whole, and then randomly samples $l = O(\lambda)$ entries of the encrypted code-words $\llbracket c_i \rrbracket$ and checks if $c' \leftarrow \text{Enc}_C(f')$ is consistent with $\sum_{i=1}^r \gamma_i c_i$ at those entries.

- **Evaluation Phase (at point $r = (r_0, \dots, r_{n-1})$).**

1. Let $q_1 = \bigotimes_{i=0}^{s-1} (1 - r_i, r_i)$ and $q_2 = \bigotimes_{i=s}^{n-1} (1 - r_i, r_i)$. $\mathcal{V}$ sends q_1 to $\mathcal{P}$.

2. $\mathcal{P}$ computes the plaintext-ciphertext multiplication $\llbracket f'' \rrbracket = \langle q_1, (\llbracket f_1 \rrbracket, \dots, \llbracket f_r \rrbracket) \rangle$ and sends to $\mathcal{V}$.
3. $\mathcal{V}$ randomly samples $l = O(\lambda)$ columns of the encrypted codewords $\llbracket c_i \rrbracket$, decrypts the random linear combination of rows $f'' \in \mathcal{F}_p^r$ and checks if $c'' \leftarrow \text{Enc}_C(f'')$ is consistent with $\sum_{i=1}^r q_{1i} c_i$ at those entries.
4. $\mathcal{V}$ outputs $\langle q_2, f'' \rangle$ as the evaluation of $f(r)$.

Complexity Analysis We analyze the complexity that the PCS framework incurs to the proof system. The detailed operation counts are provided in Table 3.

1. **Commit Phase.** $\mathcal{P}$ encodes r rows of the coefficient matrix and for each row the encoding costs T_m , incurring as total $r = O(\sqrt{N})$ encoding operations. In addition, $\mathcal{P}$ computes a Merkle Tree of a vector of length $r \times M$, operating $O(r \times M) = O(N)$ hashes. Finally, the communication of com_{MT} brings $O(1)$ hash proof size.
2. **Testing Phase.** $\mathcal{P}$ computes a random linear combination $\llbracket f' \rrbracket$ of the encrypted rows, incurring $O(N)$ plaintext-ciphertext multiplications. Then $\mathcal{V}$ receives $\llbracket f' \rrbracket$, opens $O(\lambda)$ columns of the encoded matrix, decrypts all of the entries and checks if $\sum_{i=1}^r \gamma_i c_i = c' \leftarrow \text{Enc}_C(f')$ at those entries, incurring $O(\lambda r) + T_m = O(\lambda \sqrt{N})$ plaintext-plaintext multiplications, one encoding of message length m and $O(\lambda r) = O(\lambda \sqrt{N})$ decryptions and a communication cost of $O(M + \lambda r) = O(\lambda \sqrt{N})$ ciphertexts.
3. **Evaluation Phase.** The prover's and the verifier's tasks are almost the same with those in Testing Phase. $\mathcal{P}$ computes a linear combination of r encrypted rows and sends it to $\mathcal{V}$ with weights q_1 , incurring $O(N)$ plaintext-ciphertext multiplications and $O(M) = O(\sqrt{N})$ ciphertexts. Then $\mathcal{V}$ reads and decrypts it, opens another $O(\lambda)$ columns of the encrypted encoded matrices, decrypts them and checks the equation $\sum_{i=1}^r q_{1i} f_i = c'' \leftarrow \text{Enc}_C(f'')$ at those entries, operating $O(\lambda r) = O(\lambda \sqrt{N})$ plaintext multiplications and one encoding. Finally $\mathcal{V}$ computes the inner product $\langle c'', q_2 \rangle$, operating $O(M) = O(\sqrt{N})$ plain multiplications.

6.4 Security Analysis

Field Choice Traditional PCS typically favor large prime fields as their base fields, such as Babybear ($p = 2^{31} - 2^{27} + 1$) or Goldilocks ($p = 2^{64} - 2^{32} + 1$). This preference is driven by modern hardware architectures where these fields

Table 3. Detailed Operation Counts for Our Blind PCS

Step	Level	Prover Operations	Verifier Operations	Proof Size
1. Commit	0	$\frac{2N}{B\rho}$ (pt-ct mults) $\frac{r}{B}(\frac{2m}{\rho} - 2)$ (adds)	0	0
2. Testing	1	$\frac{N}{B\rho}$ (pt-ct mults) $(\frac{r}{B} - 1)\frac{m}{\rho}$ (ct-ct adds)	$\lambda r + \frac{2m}{\rho}$ (pt-pt mults) λr (decryptions)	$M + \lambda r$ (cts)
3. Evaluation	1	$\frac{N}{B\rho}$ (pt-ct mults) $(\frac{r}{B} - 1)\frac{m}{\rho}$ (ct-ct adds)	$\lambda r + \frac{3m}{\rho}$ (pt-pt mults) λr (decryptions)	$M + \lambda r$ (cts)

benefit from specialized instruction set optimizations, significantly accelerating basic arithmetic operations.

However, in the context of FHE settings, a large plaintext field p introduces a significant ciphertext modulus expansion. For instance, when a 50-bit plaintext field is used with a SIMD Batch Size of $N = 2^{14}$, the required ciphertext modulus Q can increase to approximately 420 bits. Since the Prover must perform all computations directly on these high-dimensional ciphertext RNS limbs, the theoretical efficiency of large prime fields is eclipsed by the overhead of FHE arithmetic.

In this paper, we depart from the "large base field" paradigm and decide to adopt an FHE-friendly small field $\mathbb{F}_p$ (approximately 16 bits) for our primary interactions. A smaller plaintext modulus allows for a much tighter ciphertext modulus (e.g., ~ 150 bits), reducing the number of RNS limbs and accelerating every homomorphic operation.

To compensate for the reduced soundness of a small base field, we utilize extension field sampling for challenges. By lifting the query phase to an extension field $\mathbb{F}_{p^k}$, we ensure that the probability of a successful cheating attempt remains cryptographically negligible. Notably, the commitment and encoding phases are still performed over the base field $\mathbb{F}_p$ to minimize computational and storage overhead, whereas the elevation to the extension field $\mathbb{F}_{p^k}$ occurs only during the computation of random linear combinations to satisfy the required soundness level.

Soundness Analysis The soundness analysis of our blind PCS follows the same framework established in Brakedown [20]. The primary distinction lies in the specific properties of the underlying linear code. Specifically, our construction utilizes the Generalized RAA code instead of the RS code. This difference is encapsulated in the relative minimum distance parameter δ within the following theorem:

Theorem 9. *The blind PCS has the following bound for the soundness error:*

$$\epsilon_{total} \leq \left(1 - \frac{\delta}{3}\right)^\ell + \frac{N}{p^k}$$

where ℓ denotes the query times, δ denotes the relative minimum distance of Generalized RAA code and $\mathbb{F}_{p^k}$ is the extension field.

7 Evaluations

This section discusses the server runtime in our blind PCS presented in Section 5. Furthermore, we compare the estimated performance with those of the closest related works, *Laminate* [26] and *Phalanx* [32]. To provide a rigorous and scalable analysis, we follow the standard practice established by SOTA frameworks such as *Laminate* and *Phalanx*.

We implemented the core cryptographic primitives using the OpenFHE library, targeting the BGV homomorphic encryption scheme. Given that existing state-of-the-art works are primarily constructed based on the BFV/BGV schemes to exploit SIMD-parallelism, we likewise select the BGV scheme for our implementation. This choice ensures a fair and consistent performance comparison with these benchmarks while maintaining compatibility with the most widely adopted FHE infrastructures in current research.

7.1 Implementation Discussion

All performance evaluations and microbenchmarks were executed on a 14-inch MacBook Pro featuring the Apple M4 Pro chip and 24 GB of RAM, operating on macOS Sequoia 15.6.1. We conducted microbenchmarks using Microsoft’s SEAL library and performed performance estimations based on the analytical complexity formulas presented in the previous section. The microbenchmark of core FHE operations are presented in Table 4 results. We adopt the same parameter set as *LAMINATE* to ensure consistency: the plaintext modulus is set to 16 bits ($p = 2^{16} + 1$) with an extension degree of 8 to achieve at least 100-bit security. For the ciphertext space, we employ 3 RNS limbs, totaling 150 bits, with a ring dimension of $N = 2^{14}$.

Table 4. Microbenchmark of Core Cryptographic Operations

Operation	Latency (ms)
Homomorphic Addition	0.03
Plaintext-by-Ciphertext Multiplication	1.607
Decryption	0.99

7.2 Comparisons

As demonstrated in our complexity analysis, both Ligero and Basefold rely on RS codes. To achieve constant multiplicative depth within the FHE setting, the

encoding process for RS codes utilizes a 3-layer 2D-NTT technique. This shift causes the computational complexity to escalate from the traditional $O(n \log n)$ to $O(kn^{1+1/k})$.

Prover Time In this specific FHE setting, we observe a trend reversal compared to traditional PCS. Basefold exhibits slower performance than Ligerio. This is primarily because Basefold encodes the entire polynomial simultaneously. In contrast, Ligerio partitions the polynomial into a matrix and encodes each row individually. In a traditional setting, Basefold’s encoding time is $O(n \log n)$ and Ligerio’s is $\sqrt{n} \times O(\sqrt{n} \log \sqrt{n}) = O(n \log n)$; Basefold typically performs faster due to superior constant factors. However, in the FHE setting, Basefold’s complexity becomes $O(kn^{1+1/k})$, while Ligerio’s complexity is $\sqrt{n} \times O(k\sqrt{n}^{1+1/k}) = O(kn^{1+1/2k})$. Because Ligerio has a smaller exponent in this setting, it delivers better performance. Notably, our RAA-based PCS maintains a strictly linear complexity of $O(n)$, showcasing a significant performance advantage over both RS-code-based schemes as the circuit scale N increases. The empirical results of prover complexity across varying circuit scales are illustrated in Figure 5. In addition, we provide the table of raw data in Table 5. The ratios provide a concrete metric for the "complexity-depth" solution. At a scale of 2^{16} , our prover is roughly $8.5\times$ faster than Basefold. Further, by 2^{30} , our prover is approximately $193\times$ faster.

Proof Size At smaller circuit scales, our proof size is comparable to that of Basefold, a result primarily attributed to our selection of a 16-bit small field. Although a 16-bit base field necessitates a higher degree of field extension for security, the encoding process is executed entirely within the base field. Consequently, the ciphertexts committed to the Merkle tree remain in the base field; transmission in the extension field is only required for the rows generated after the random linear combination. This small field strategy is highly optimized for proof size, demonstrating a distinct advantage over Ligerio. The empirical results of proof size are illustrated in Figure 6.

Table 5. Benchmark Comparison of estimated Prover Time (ms) and Ratios. Our scheme maintains strictly linear growth, delivering significant speedups over Phalanx and LAMINATE as the circuit scale N increases.

Size (2^n)	16	18	20	22	24	26	28	30
Ours (ms)	102.4	409.6	1638.4	6553.6	26214.4	104857.6	419430.4	1677721.6
Phalanx (ms)	824	4105.6	20480	102361	512459.2	2569011.2	12892495.76	64756069.58
LAMINATE (ms)	875.2	5376	33257.6	207179.6	1297612.8	8160043.2	51465125.4	325268275.2
<i>Phalanx/Ours</i>	8.05 $\times$	10.02 $\times$	12.50 $\times$	15.62 $\times$	19.55 $\times$	24.50 $\times$	30.74 $\times$	38.60 $\times$
<i>LAMINATE/Ours</i>	8.55 $\times$	13.13 $\times$	20.30 $\times$	31.61 $\times$	49.50 $\times$	77.82 $\times$	122.70 $\times$	193.88 $\times$

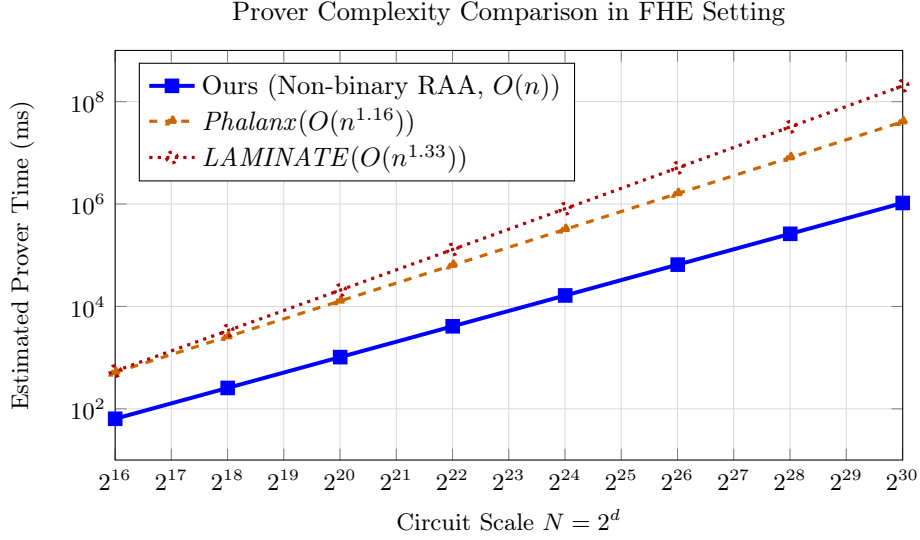


Fig. 5. Prover Complexity Comparison in FHE Setting. This chart compares the normalized prover complexity across varying circuit scales ($N = 2^{16}$ to 2^{30}).

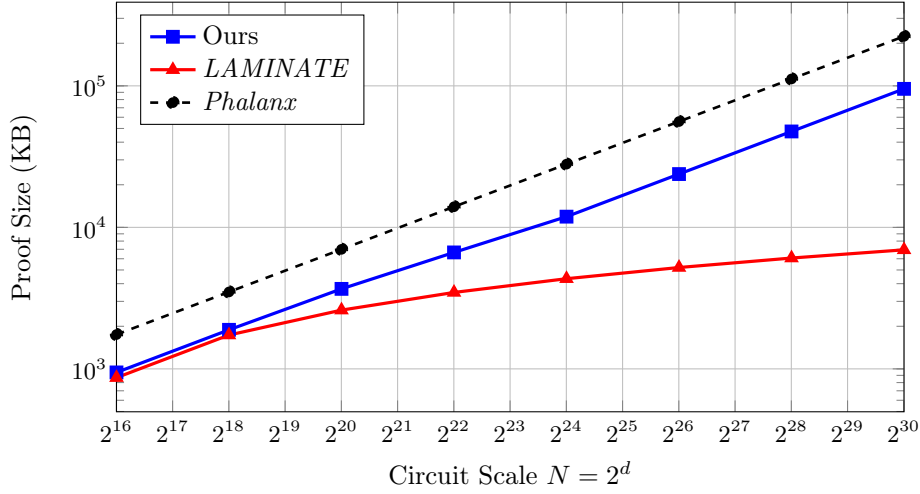


Fig. 6. Proof Size Comparison. This chart illustrates the growth of proof size (in KB) relative to the circuit scale. By utilizing a 16-bit plaintext field, our scheme achieves a compact 150-bit ciphertext configuration, resulting in a proof size comparable to Basefold at smaller scales ($N \leq 2^{20}$).

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A Deferred Proofs

A.1 Proof of Theorem 5

Proof. For the ease of presenting, we alias $\lfloor \frac{w-i}{2} \rfloor$ as a and $\lceil \frac{w-i}{2} \rceil$ as b . We have $b - a \in \{0, 1\}$. First we analyze the dominant term in $A(w, h)$ where $h \in [\delta N]$.

$$A(w, h) = \sum_{i=0}^{w-1} \frac{\binom{h-1}{a-1} \binom{N-h}{b} \binom{h-a}{i}}{\binom{N}{w}} \cdot \frac{(q-1)^a (q-2)^i}{(q-1)^w} \quad (6)$$

$$\leq \sum_{i=0}^{w-1} \frac{\binom{h}{a} \binom{N-h}{b} \binom{h-a}{i}}{\binom{N}{w}} \quad (7)$$

First we consider the case where $a = b$. We write the parameters as follows:

$$a = b = \alpha(N)N, h = \beta(N)N, i = \theta(N)N$$

We emphasize that $\alpha(N), \beta(N), \theta(N)$ may relate to N and are not constants. For the ease of presenting, we write α, β, θ in the following analysis. Since $w = a + b + i$, we have $w = (2\alpha + \theta)N$.

Using Stirling formula, we have the following estimations:

$$\begin{aligned} \binom{h}{a} &\leq e^{H(\frac{\alpha}{\beta})\beta N} \\ \binom{N-h}{b} &\leq e^{H(\frac{\alpha}{1-\beta})(1-\beta)N} \\ \binom{h-a}{i} &\leq e^{H(\frac{\theta}{\beta-\alpha})(\beta-\alpha)N} \\ \binom{N}{w} &\geq \frac{1}{N} \cdot e^{H(2\alpha+\theta)N} \end{aligned}$$

We use the above estimations to replace the combinatorial terms in expression 7 and derive

$$A(w, h) \leq wN \cdot \max_{0 \leq i \leq w-1} e^{\Phi(\alpha, \beta, \theta)N}$$

where $\Phi(\alpha, \beta, \theta) = (1 - \beta)H(\frac{\alpha}{1-\beta}) + \beta H(\frac{\alpha}{\beta}) + (\beta - \alpha)H(\frac{\theta}{\beta-\alpha}) - H(2\alpha + \theta)$

Note that

$$\begin{aligned} D(w) &= \sum_{h=1}^{\delta N} A(w, h) \\ &\leq \delta N \cdot \max_{1 \leq h \leq \delta N} A(w, h) \\ &\leq \delta w N^2 \max_{\substack{0 \leq i \leq w-1 \\ 1 \leq h \leq \delta N}} e^{\Phi(\alpha, \beta, \theta)N} \end{aligned}$$

Now the central task is to analyze the maximum point of $\Phi(\alpha, \beta, \theta)$ given the constraints

$$\begin{aligned} 2\alpha + \theta &= \frac{w}{N} \geq 0.01 \\ \alpha + \theta &\leq \frac{h}{N} = \beta \leq \delta \end{aligned}$$

We give some essential results on the derivatives of $H(x)$ and $f(x) \triangleq xH(\frac{\alpha}{x})$.

$$H'(x) = \ln\left(\frac{1-x}{x}\right) \quad (8)$$

$$f'(x) = -\ln\left(1 - \frac{\alpha}{x}\right) \quad (9)$$

$$f''(x) = -\frac{\alpha}{x(x-\alpha)} \quad (10)$$

Therefore, we have

$$\begin{aligned} \frac{\partial}{\partial \beta} \Phi(\alpha, \beta, \theta) &= \log\left(\frac{1-\alpha-\beta}{1-\beta}\right) - \log\left(\frac{\beta-\alpha}{\beta}\right) - \log\left(\frac{\beta-\alpha-\theta}{\beta-\alpha}\right) \\ &= \log\left(\frac{(1-\beta-\alpha)\beta}{(1-\beta)(\beta-\alpha-\theta)}\right) \end{aligned}$$

Note that

$$\begin{aligned} \beta \leq \delta < \frac{1}{2} < \frac{\alpha + \theta}{2\alpha + \theta} &\iff \\ (1-\alpha)\beta > (1+\alpha+\theta)\beta - (\alpha + \theta) &\iff \\ (1-\beta-\alpha)\beta > (1-\beta)(\beta-\alpha-\theta) \end{aligned}$$

Therefore we have $\frac{\partial}{\partial \beta} \Phi(\alpha, \beta, \theta) > 0$ when $\beta \leq \delta < \frac{1}{2}$. And $\max \Phi(\alpha, \beta, \theta) = \Phi(\alpha, \delta, \theta) \triangleq \Phi(\alpha, \theta)$. Now we fix α and analyze the monotonicity of $\Phi(\alpha, \theta)$ in θ . We emphasize that $w = 2\alpha + \theta$ is a function of θ now.

$$\frac{\partial}{\partial \theta} \Phi(\alpha, \theta) = \log\left(\frac{(\delta - \alpha - \theta)(2\alpha + \theta)}{\theta(1 - 2\alpha - \theta)}\right)$$

$\frac{\partial}{\partial \theta} \Phi(\alpha, \theta) = 0$ when $\theta = \frac{2\alpha(\delta - \alpha)}{1 + \alpha - \delta}$. Therefore

$$\begin{aligned} \max_{\theta} \Phi(\alpha, \theta) &= \Phi\left(\alpha, \frac{2\alpha(\delta - \alpha)}{1 + \alpha - \delta}\right) \\ &= \delta H\left(\frac{\alpha}{\delta}\right) + (1 - \delta)H\left(\frac{\alpha}{1 - \delta}\right) - (1 - \delta + \alpha)H\left(\frac{2\alpha}{1 - \delta + \alpha}\right) \\ &\leq \delta H\left(\frac{\alpha}{\delta}\right) + (1 - \delta)H\left(\frac{\alpha}{1 - \delta}\right) - (1 - \delta)H\left(\frac{2\alpha}{1 - \delta}\right) \triangleq \Psi(\alpha) \end{aligned}$$

Furthermore,

$$\begin{aligned}
\frac{d}{d\alpha}\Psi(\alpha) &= \log\left(\frac{\delta - \alpha}{\alpha}\right) + \log\left(\frac{1 - \delta - \alpha}{\alpha}\right) - 2\log\left(\frac{1 - \delta - 2\alpha}{2\alpha}\right) \\
&= \log\left(\frac{4(\delta - \alpha)(1 - \delta - \alpha)}{(1 - \delta - 2\alpha)^2}\right) \\
&= \log\left(1 + \frac{(5\delta - 1)(1 - \delta) - 4\delta\alpha}{(1 - \delta - 2\alpha)^2}\right)
\end{aligned}$$

Therefore, when $\delta < 0.2$, $\frac{d}{d\alpha}\Psi(\alpha) < 0$ for all α . Note that $2\alpha + \theta \geq 0.01$ and $\Phi(\alpha, \beta, \theta)$ achieves its maximum at $\theta = \frac{2\alpha(\delta - \alpha)}{1 + \alpha - \delta}$. Therefore we have $\frac{2\alpha}{1 - \delta + \alpha} \geq 0.01$, and thus $\alpha \geq \frac{1 - \delta}{199}$, $\max \Psi(\alpha) = \Psi(\frac{1 - \delta}{199}) < 0$ when $\delta < 0.2$.

Overall, we have proven that when $w \geq 0.01N$, $\Phi(\alpha, \beta, \theta) \leq \Psi(\frac{1 - \delta}{199}) < 0$ and

$$\begin{aligned}
D(w) &\leq \delta w N^2 \max_{\substack{0 \leq i \leq w-1 \\ 1 \leq h \leq \delta N}} e^{\Phi(\alpha, \beta, \theta)N} \\
&\leq \delta w N^2 e^{\Psi(\frac{1 - \delta}{199}) \frac{N}{w} \cdot w} \\
&\leq e^{(100\Psi(\frac{1 - \delta}{199}) + \epsilon)w}
\end{aligned}$$

where $\Psi(\frac{1 - \delta}{199}) < 0$. Thus for all $0 < \Lambda < -100\Psi(\frac{1 - \delta}{199})$, we have $D(w) < e^{-\Lambda w}$, as desired. $\square$

A.2 Proof of Corollary 1

Proof. By the Stirling's Formula, we have

$$\left(\frac{N}{k}\right) \leq e^{H(\frac{kw}{N}) \cdot \frac{N}{k}}$$

. Now take the logarithm of the left side, we have

$$\begin{aligned}
\log\left(\left(\frac{N}{k}\right)(q - 1)^w D(kw)\right) &= \log\left(\left(\frac{N}{k}\right)\right) + w \cdot \log(q - 1) + \log(D(kw)) \\
&\leq H\left(\frac{kw}{N}\right) \cdot \frac{N}{k} + (\log(q - 1) - k\Lambda) \cdot w \\
&\triangleq f(w)
\end{aligned}$$

Use the equation 8 and we take the derivative of $f(w)$ in the last line of the above analysis:

$$f'(w) = \log\left(\frac{N - kw}{kw}\right) + \log(q - 1) - k\Lambda$$

Note that $w \geq 0.01N$, therefore we have the first term $\log(\frac{N - kw}{kw}) \leq \log(\frac{100 - k}{k}) < 5$. Now if we restrict $k > \frac{5 + \log(q - 1)}{\Lambda}$, we can have

$$f'(w) < 5 + \log(q - 1) - \frac{5 + \log(q - 1)}{\Lambda} \cdot \Lambda = 0$$

Therefore, in such case $f(w)$ achieves its maximum at $w = 0.01N$, which is

$$f(w) \leq (H(0.01k) \cdot \frac{1}{k} + 0.01 \log(q-1) - 0.01k\Lambda) \cdot N$$

Therefore, if $k > \frac{100H(0.01) + \log(q-1)}{\Lambda}$, we have the following:

$$f(w) \leq (H(0.01k) \cdot \frac{1}{k} + 0.01 \log(q-1) - 0.01k\Lambda) \cdot N \quad (11)$$

$$\leq 0.01N \cdot (100H(0.01) + \log(q-1) - k\Lambda) < 0 \quad (12)$$

To conclude, for $K > \frac{100H(0.01) + \log(q-1)}{\Lambda}$, we can find $C(K, q) = K\Lambda - (100H(0.01) + \log(q-1)) > 0$ such that for all $k > K$, the following holds

$$\left(\frac{N}{k}\right) (q-1)^w D(kw) = o(e^{-CN})$$

□

A.3 Proof of Corollary 2

Proof. By the Stirling's Formula, we have

$$\left(\frac{N}{k}\right) \leq e^{H(\frac{kw}{N}) \cdot \frac{N}{k}}$$

Now note that when $w \leq \frac{0.01N}{k}$, the right side $e^{H(\frac{kw}{N}) \cdot \frac{N}{k}}$ is increasing in w . Therefore, we have

$$\left(\frac{N}{k}\right) (q-1)^w \leq e^{H(\frac{kw}{N}) \cdot \frac{N}{k}} \cdot (q-1)^w \leq e^{H(0.01) \cdot \frac{N}{k}} \cdot (q-1)^{0.01N}$$

Now by Theorem 5, we have

$$\begin{aligned} \sum_{s=0.01N}^{\delta N} D(s) &\leq \sum_{s=0.01N}^{\delta N} e^{-\Lambda s} \\ &< (\delta - 0.01)N e^{-\Lambda 0.01N} \\ &< N \cdot e^{-\Lambda \cdot 0.01N} \end{aligned}$$

Now, combine the above estimations of the former and latter terms and we derive

$$\begin{aligned} &\log\left(\left(\frac{N}{k}\right) (q-1)^w \sum_{s=0.1N}^{\delta N} D(s)\right) \\ &\leq H(0.01) \cdot \frac{N}{k} + \frac{0.01N}{k} \cdot \log(q-1) + \log N - \Lambda \cdot 0.01N \\ &= \frac{0.01N}{k} \cdot (100H(0.01) + \log(q-1) + \frac{\log N}{N} - k\Lambda) \end{aligned}$$

Note that the last expression is almost the same as the expression 11 except for an extra term $\frac{\log N}{N}$ which tends to 0 when $N \rightarrow \infty$, therefore we have proved that for $K > \frac{100H(0.01)+\log(q-1)}{\Lambda}$, we can find $C(K, q) = K\Lambda - (100H(0.01) + \log(q-1)) > 0$ such that for all $k > K$, the following holds

$$\binom{\frac{N}{k}}{w} (q-1)^w \sum_{s=0.1N}^{\delta N} D(s) = o(e^{-CN})$$

for any $k > K, w \leq \frac{0.01N}{k}$. $\square$

A.4 Proof of Lemma 4

Proof. For the ease of presenting, we alias $\lfloor \frac{w-i}{2} \rfloor = a, \lceil \frac{w-i}{2} \rceil = b$, and we have $a+b=w$ and $a-b \in \{0,1\}$. Instead of using the Stirling formula directly, we have to apply some fine-grained estimations of the combinatorial terms in $A(w, h)$:

$$\frac{\binom{h-1}{a-1} \binom{N-h}{b} \binom{h-a}{i}}{\binom{N}{w}} \cdot \frac{(q-1)^a (q-2)^i}{(q-1)^w}$$

Note that $a+i \leq \frac{w+i+1}{2} \leq w$, we have the latter term $\frac{(q-1)^a (q-2)^i}{(q-1)^w} \leq 1$.

For the former term, first we note that

$$\binom{N}{w} = \frac{N}{w} \cdot \binom{N-1}{w-1}$$

We have the following:

$$\frac{\binom{h-1}{a-1} \binom{N-h}{b} \binom{h-a}{i}}{\binom{N}{w}} \cdot \frac{(q-1)^a (q-2)^i}{(q-1)^w} \leq \frac{w}{N} \cdot \frac{\binom{h-1}{a-1} \binom{N-h}{b} \binom{h-a}{i}}{\binom{N-1}{w-1}}$$

Now consider the following:

$$\binom{h-1}{i} \binom{N-h}{j} = \frac{(h-1) \cdots (h-i)}{i!} \cdot \frac{(N-h)(N-h-1) \cdots (N-h-j+1)}{j!}$$

Note that the function $(h-x)(N-h-j+x)$ is decreasing in x when $x \geq 0$, and we have $(h-x)(N-h-j+x) \leq h(N-h-j)$.

When $a = b+1$, we have

$$\begin{aligned} \binom{h-1}{a-1} \binom{N-h}{b} &= \frac{(h-1) \cdots (h-a+1)}{(a-1)!} \cdot \frac{(N-h)(N-h-1) \cdots (N-h-b+1)}{b!} \\ &\leq \frac{(h(N-h-b+1))^b}{(b!)^2} \end{aligned}$$

Furthermore, with the estimations $\binom{h-a}{i} = \frac{(h-a)(h-a-1)\cdots(h-a-i+1)}{i!} \leq \frac{h^i}{i!}$ and $\binom{N-1}{w-1} \geq \frac{(N-w)^{w-1}}{(w-1)!}$, we can derive the following:

$$\begin{aligned} \frac{\binom{h-1}{a-1} \binom{N-h}{b} \binom{h-a}{i}}{\binom{N}{w}} &\leq \frac{w}{N} \cdot \binom{w-1}{a-1, b, i} \cdot \frac{h^{a+i} (N-h-a+1)^b}{(N-w)^w} \\ &\leq \frac{w}{N} \cdot \binom{w-1}{a-1, b, i} \cdot \left(\frac{h}{N-w}\right)^{a+i} \\ &\leq \frac{w \cdot 3^w}{N} \cdot \left(\frac{h}{N-w}\right)^{\frac{w}{2}} \end{aligned}$$

When $a = b$, we have

$$\begin{aligned} &\binom{h-1}{a-1} \binom{N-h}{b} \\ &= \frac{(h-1) \cdots (h-a+1)}{(a-1)!} \cdot \frac{(N-h)(N-h-1) \cdots (N-h-b+1)}{b!} \\ &\leq \frac{(h(N-h-a+1))^{a-1}}{(a-1)!b!} \cdot (N-h-b+1) = \frac{h^{a-1} (N-h-b+1)^b}{(a-1)!b!} \end{aligned}$$

The rest analysis is exactly the same with the case $a = b + 1$.

Combining the analysis altogether, we can derive the following:

$$\begin{aligned} A(w, h) &= \sum_{i=0}^{w-1} \frac{\binom{h-1}{a-1} \binom{N-h}{b} \binom{h-a}{i}}{\binom{N}{w}} \cdot \frac{(q-1)^a (q-2)^i}{(q-1)^w} \\ &\leq w \cdot \left(\frac{w \cdot 3^w}{N} \cdot \left(\frac{h}{N-w}\right)^{\frac{w}{2}}\right) = \frac{w^2}{N} \cdot \left(\frac{h}{N-w}\right)^{\frac{w}{2}} \cdot 3^w \end{aligned}$$

For the second statement, we note that

$$\sum_{i=1}^{\delta N} i^{\frac{w}{2}} \leq \int_1^{\delta N+1} x^{\frac{w}{2}} dx = \frac{2}{w+2} \cdot (\delta N+1)^{\frac{w}{2}+1}$$

Therefore, we have

$$\begin{aligned} D(w) &= \frac{w^2}{N} \cdot \left(\frac{1}{N-w}\right)^{\frac{w}{2}} \cdot 3^w \cdot \sum_{h=1}^{\delta N} h^{\frac{w}{2}} \\ &\leq \frac{2w^2}{w+1} \cdot \frac{\delta N+1}{N} \cdot \left(\frac{\delta N+1}{N-w}\right)^{\frac{w}{2}} \cdot 3^w \end{aligned}$$

□

A.5 Proof of Lemma 5

Proof. We are now taking a closer look at the most complicated term $A(kw, s) \cdot D(s)$. We use the estimations provided in the Lemma 4

$$\begin{aligned} A(kw, s) \cdot D(s) &\leq \frac{(kw)^2}{N} \cdot \left(\frac{s}{N-kw}\right)^{\frac{kw}{2}} \cdot 3^w \cdot \frac{2s^2}{s+1} \cdot \frac{\delta N+1}{N} \cdot \left(\frac{\delta N+1}{N-s}\right)^{\frac{s}{2}} \cdot 3^s \\ &\leq \frac{(kw)^2}{N} \cdot \left(\frac{s}{N-kw}\right)^{\frac{kw}{2}} \cdot 3^w \cdot 2s \cdot \frac{\delta N+1}{N} \cdot \left(\frac{\delta N+1}{N-0.01N}\right)^{\frac{s}{2}} \cdot 3^s \\ &\triangleq g(s; w, k, N) \end{aligned}$$

We omit the notation of w, k, N of g in the following proof. We take the logarithm and then take the derivative of g :

$$\frac{d}{ds} \log g(s) = \frac{kw}{2s} + \frac{2}{s} + \frac{1}{2} \cdot \log\left(\frac{\delta N+1}{0.99N}\right) + \log 3$$

Now note that $s \geq \frac{kw}{2}$ (otherwise $A(kw, s) = 0$), we can choose $\delta < 0.99 \cdot \left(\frac{1}{3e^2}\right)^2$ such that

$$\frac{d}{ds} \log g(s) \leq 1 + 1 + \frac{1}{2} \cdot \log\left(\frac{\delta N+1}{0.99N}\right) + \log 3 < 0$$

In such case, g achieves its maximum at $s = kw/2$ and the value is

$$\begin{aligned} g(s) &\leq g\left(\frac{kw}{2}\right) = \frac{(kw)^2}{N} \cdot \left(\frac{kw}{2(N-kw)}\right)^{\frac{kw}{2}} \cdot 3^w \cdot kw \cdot \frac{\delta N+1}{N} \cdot \left(\frac{\delta N+1}{N-0.01N}\right)^{\frac{kw}{4}} \cdot 3^{\frac{kw}{2}} \\ &= (kw)^3 \cdot \frac{\delta N+1}{N^2} \cdot 3^{w+\frac{kw}{2}} \cdot \left(\frac{kw}{2(N-kw)}\right)^{\frac{kw}{2}} \cdot \left(\frac{\delta N+1}{0.99N}\right)^{\frac{kw}{4}} \\ &\leq \frac{(kw)^3 \cdot \delta'}{N} \cdot 3^{(1+\frac{k}{2})w} \cdot \left(\frac{kw}{2 \cdot 0.99N}\right)^{\frac{kw}{2}} \cdot \left(\frac{\delta'}{0.99}\right)^{\frac{kw}{4}} \end{aligned}$$

where δ' is any real number that is greater than δ .

Combining the Stirling Formula $\left(\frac{N}{k}\right) \leq \left(\frac{Ne}{kw}\right)^w$, we have

$$\begin{aligned} &\left(\frac{N}{k}\right) (q-1)^w \sum_{s=1}^{0.01N} A(kw, s) \cdot D(s) \\ &\leq \left(\frac{N}{k}\right) (q-1)^w \cdot 0.01N \max_{1 \leq s \leq 0.01N} g(s) \\ &\leq \left(\frac{Ne}{kw}\right)^w (q-1)^w \cdot 0.01 \cdot (kw)^3 \cdot \delta' \cdot 3^{(1+\frac{k}{2})w} \cdot \left(\frac{kw}{2 \cdot 0.99N}\right)^{\frac{kw}{2}} \cdot \left(\frac{\delta'}{0.99}\right)^{\frac{kw}{4}} \\ &= C_1 \cdot C_2^w \cdot N^{w-\frac{kw}{2}} \cdot w^{-w+3+\frac{kw}{2}} \\ &= C_1 \cdot C_2^w \cdot w^3 \cdot \left(\frac{w}{N}\right)^{(\frac{k}{2}-1)w} \triangleq p(w) \end{aligned}$$

where $C_1 = 0.01\delta' \cdot k^3$, and $C_2 = \frac{e}{k} \cdot 3^{(1+\frac{k}{2})} \cdot \left(\frac{k}{1.98}\right)^{\frac{k}{2}} \cdot \left(\frac{\delta'}{0.99}\right)^{\frac{k}{4}}$ are two constants that are irrelevant to N, w .

Now we analyze $\frac{d}{dw} \log p(w)$:

$$\begin{aligned}\frac{d}{dw} \log p(w) &= \log C_2 + \frac{3}{w} + \left(\frac{k}{2} - 1\right) \log \frac{w}{N} + \left(\frac{k}{2} - 1\right) \\ \frac{d^2}{dw^2} \log p(w) &= \frac{1}{w} \cdot \left(\frac{k}{2} - 1\right) - \frac{3}{w^2}\end{aligned}$$

Since N goes to ∞ , the derivative is negative when $w = o(N)$. In addition, $\frac{d^2}{dw^2} \log p(w) > 0$ when $w > \frac{6}{k-2}$. Therefore, $\frac{d}{dw} \log p(w)$ has at most one zero point. Therefore, the maximum of $p(w)$ is at the point $w = 1$ or $w = \frac{0.01N}{k}$.

We are analyzing $p(w)$ at both points.

$$\log p(1) = \log C_1 + \log C_2 + \left(\frac{k}{2} - 1\right) \log \frac{1}{N}$$

Now if we take $k \geq 7$, we have $p(1) = C_1 C_2 \cdot \left(\frac{1}{N}\right)^{2.5} = o\left(\frac{1}{N^2}\right)$ as desired.

$$\begin{aligned}\log p\left(\frac{0.01N}{k}\right) &= \log C_1 + \frac{0.01N}{k} \log C_2 + 3 \log \frac{0.01N}{k} + \frac{0.01N}{k} \cdot \left(\frac{k}{2} - 1\right) \log \frac{0.01}{k} \\ &= o(N) + \frac{0.01N}{k} \cdot (\log C_2 - \left(\frac{k}{2} - 1\right) \log(100k)) \\ &= \frac{k}{2} \cdot (\log 3 + \log \frac{k}{1.98} - \log(100k) - \frac{1}{2} \log(\frac{0.99}{\delta'})) + \log 3 + \log(100k)\end{aligned}$$

The last expression goes to negative for any $\delta' < 0.1$ and $k > 7$, therefore $p\left(\frac{0.01N}{k}\right) = o\left(\frac{1}{N^2}\right)$ as desired.

To conclude, we have

$$\begin{aligned}\left(\frac{N}{k}\right) (q-1)^w \sum_{s=1}^{0.01N} A(kw, s) \cdot D(s) &\leq \left(\frac{N}{k}\right) (q-1)^w \cdot 0.01N \cdot \max_{1 \leq s \leq 0.01N} g(s) \\ &\leq p(w) \leq \max(p(1), p\left(\frac{0.01N}{k}\right)) = o\left(\frac{1}{N^2}\right)\end{aligned}$$

□

A.6 Proof of Theorem 6

Proof. Note that

$$\begin{aligned}&\left(\frac{N}{w}\right) (q-1)^w \sum_{s=1}^N A(kw, s) \cdot D(s) \\ &= \sum_{s=1}^{0.01N} \left(\frac{N}{w}\right) (q-1)^w A(kw, s) \cdot D(s) + \sum_{s=0.01N}^N \left(\frac{N}{w}\right) (q-1)^w A(kw, s) \cdot D(s) \\ &\leq \sum_{s=1}^{0.01N} \left(\frac{N}{w}\right) (q-1)^w A(kw, s) \cdot D(s) + \sum_{s=0.01N}^N \left(\frac{N}{w}\right) (q-1)^w D(s)\end{aligned}$$

By Theorem 5, we have the latter term is bounded by $o(e^{-CN})$, and by Lemma 5, we have the former term is bounded by $o(\frac{1}{N^2})$.

Therefore,

$$\begin{aligned}
& \binom{\frac{N}{k}}{w} (q-1)^w \sum_{s=1}^N A(kw, s) \cdot D(s) \\
& \leq \sum_{s=1}^{0.01N} \binom{\frac{N}{k}}{w} (q-1)^w A(kw, s) \cdot D(s) + \sum_{s=0.01N}^N \binom{\frac{N}{k}}{w} (q-1)^w D(s) \\
& = o(\frac{1}{N^2}) + o(e^{-CN}) = o(\frac{1}{N^2})
\end{aligned}$$

as desired. □